香港科技大学(广州)
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

UCUG 1903 Signals and Information Technology



2026-27 Fall Term

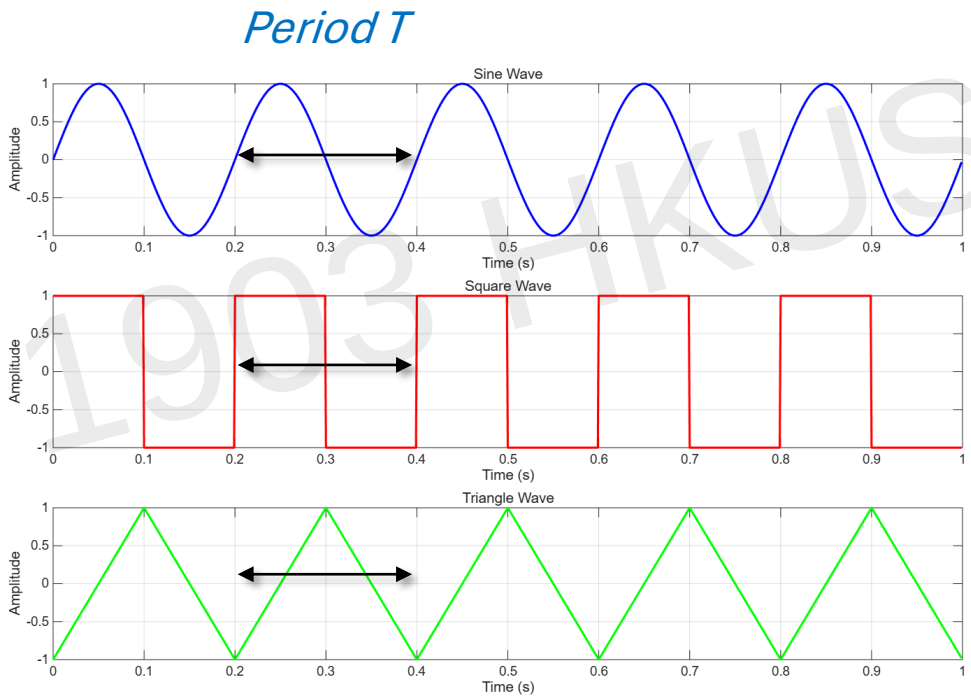
Lecture 2 Sound, Frequency and Harmonics

Dr. Yeyu TONG

yeyutong@hkust-gz.edu.cn

Periodic Signal

- A signal that oscillates with a repetitive pattern is called a *periodic signal*.
- *Period*: The *repeating interval* of a periodic signal, often notated by *T*, as shown below:



- You typically find a repeating interval from a fraction of 1 ms to several ms in the sound signal we hear.

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Fundamental Frequency

- Engineers also describe a periodic signal by its ***fundamental frequency***, ***f***, which is the reciprocal of the *period T*:

$$f = 1/T$$

- The fundamental frequency is the *repetition rate* of the signal.

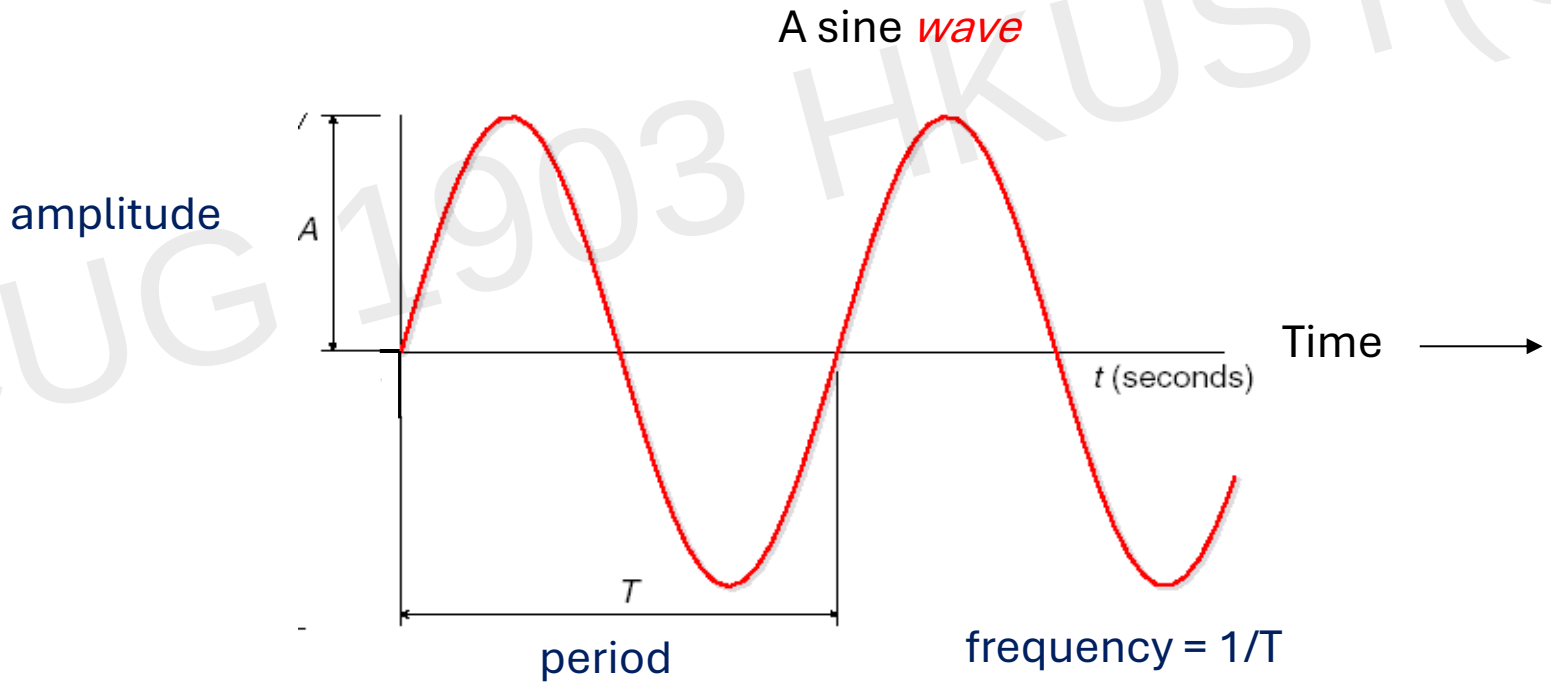
- Because *period T* has units of time (*seconds*), the unit of *frequency f* is *1/seconds* (or Hertz (Hz)).
  - e.g. A period $T = 1$ ms gives a fundamental frequency $f = 1$ *k*Hz

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A Simple Periodic Signal – Sinusoidal Signal

- We will now illustrate the perception of pitch by the simplest periodic signal - the *sinusoidal signal*. (e.g. sound of a tuning fork, or a person *whistling*). A sinusoidal signal is also called a *sine wave*.

- A sinusoidal signal that repeats with period *T* is shown below:



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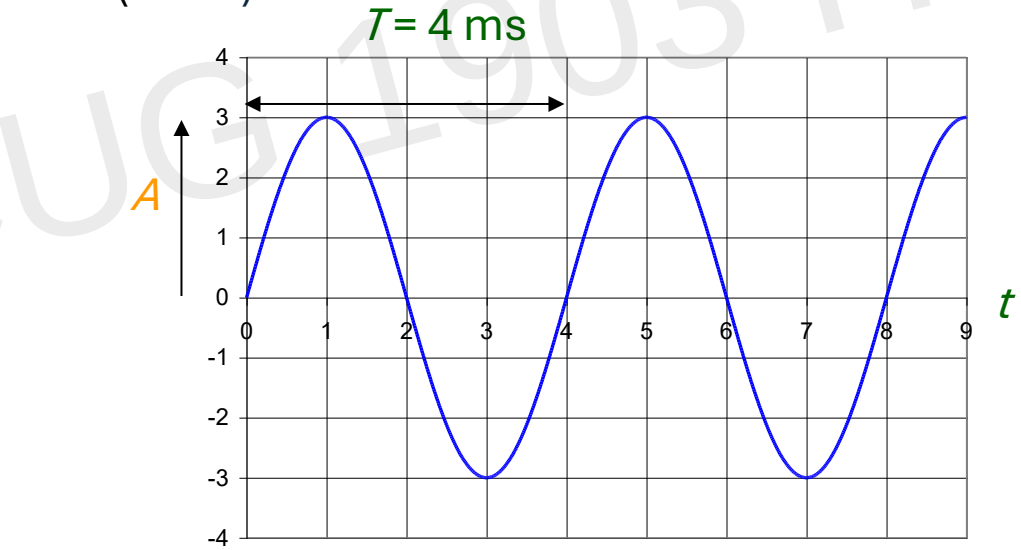
Mathematics: Sinusoidal Function in Time

Mathematics of a sine *wave*:

$$A \sin(2\pi \cdot f \cdot t)$$

- A: signal amplitude (how large a signal is)
- $\sin$: sinusoidal function (the shape of the periodic signal)
- f: signal frequency $f = 1/T$ (how frequent in time the periodic signal oscillates)
- t: time

$A \sin(2\pi f t)$



1. *amplitude* $A = 3$
2. *period* $T = 4$ ms
3. *frequency* $f = 250$ Hz

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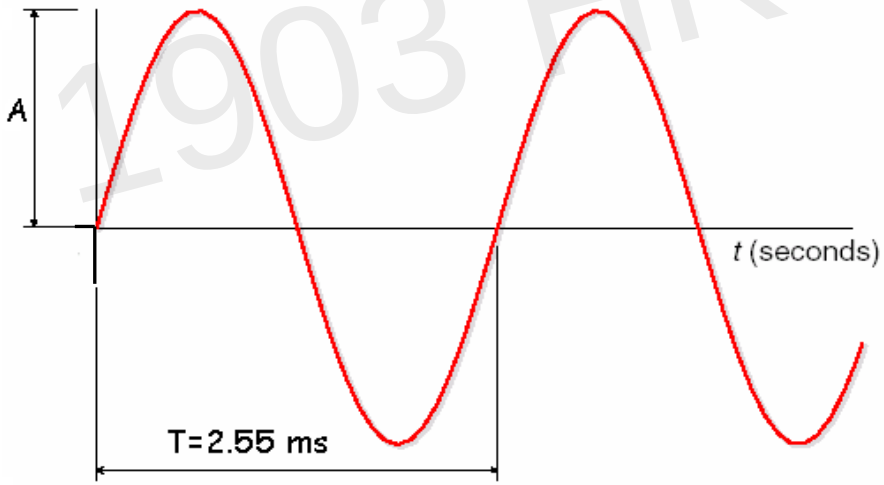
Simple example on sinusoidal signal

- Suppose a *sinusoidal signal* has

  Period T = 2.55 ms = 2.55 × 0.001 s

- then,

  Frequency f = 1/T = 1/(2.55 x 0.001 s) = 392 Hz

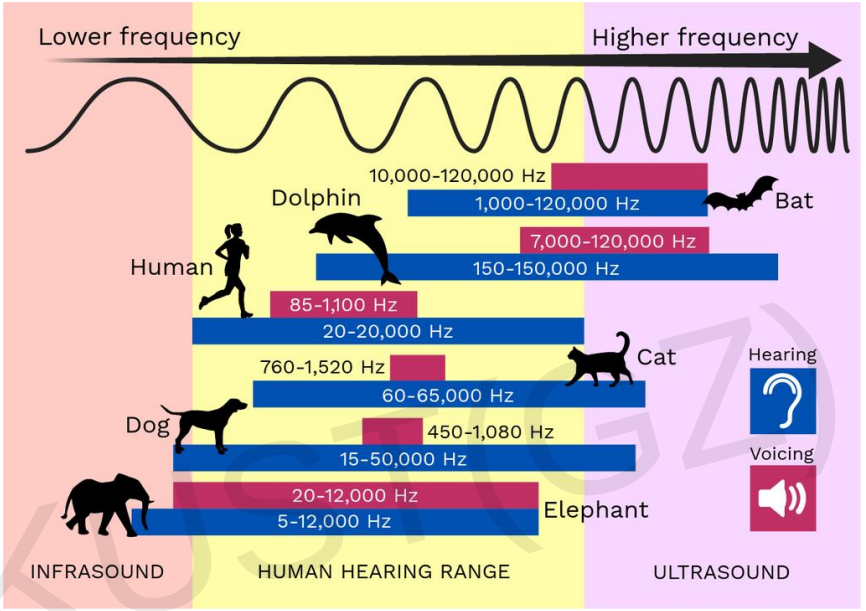


392 Hz is in fact the G note in music, or a “So” on the C scale

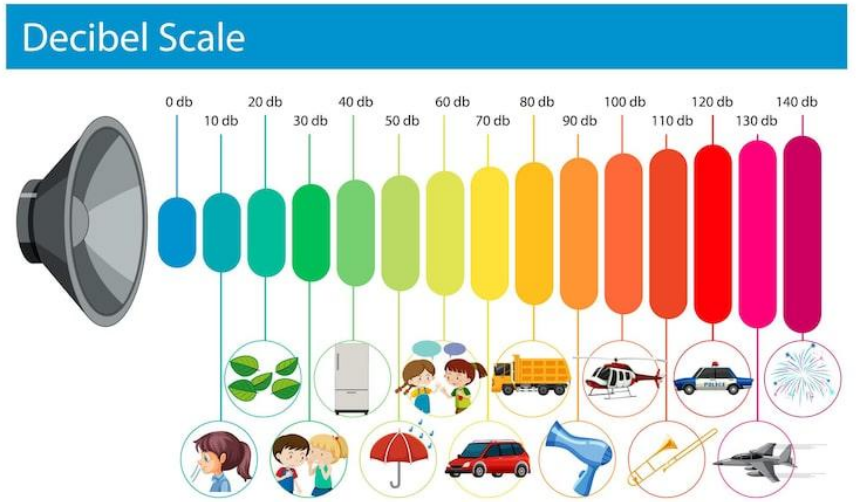
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What is Sound Signal?

- What is sound? What is in the sound signal that determines what we hear?
- Sound is an audio *signal* that we hear through variations in the air pressure that reaches our ears.



- Atmospheric pressure is ~100,000 Pa (Pascal: measure of pressure, 1 Newton per square meter). Variation of as little as 1 Pa would be a deafening sound to our ear!
- The range of frequency we can hear is from 20Hz to 20 kHz.



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A Deeper Look Into 1 Pa

- The formula for sound pressure level is $L_p = 20 \log_{10}\left(\frac{p}{p_0}\right)\ \text{dB}$ with $p_0 = 20\ \mu\text{Pa}$.
- For $p = 1\ \text{Pa}$:

$$L_p = 20 \log_{10}\left(\frac{1}{2 \times 10^{-5}}\right) = 20 \log_{10}(5 \times 10^4)$$

$$\log_{10}(5 \times 10^4) = \log_{10}(5) + \log_{10}(10^4) \approx 0.69897 + 4 = 4.69897$$

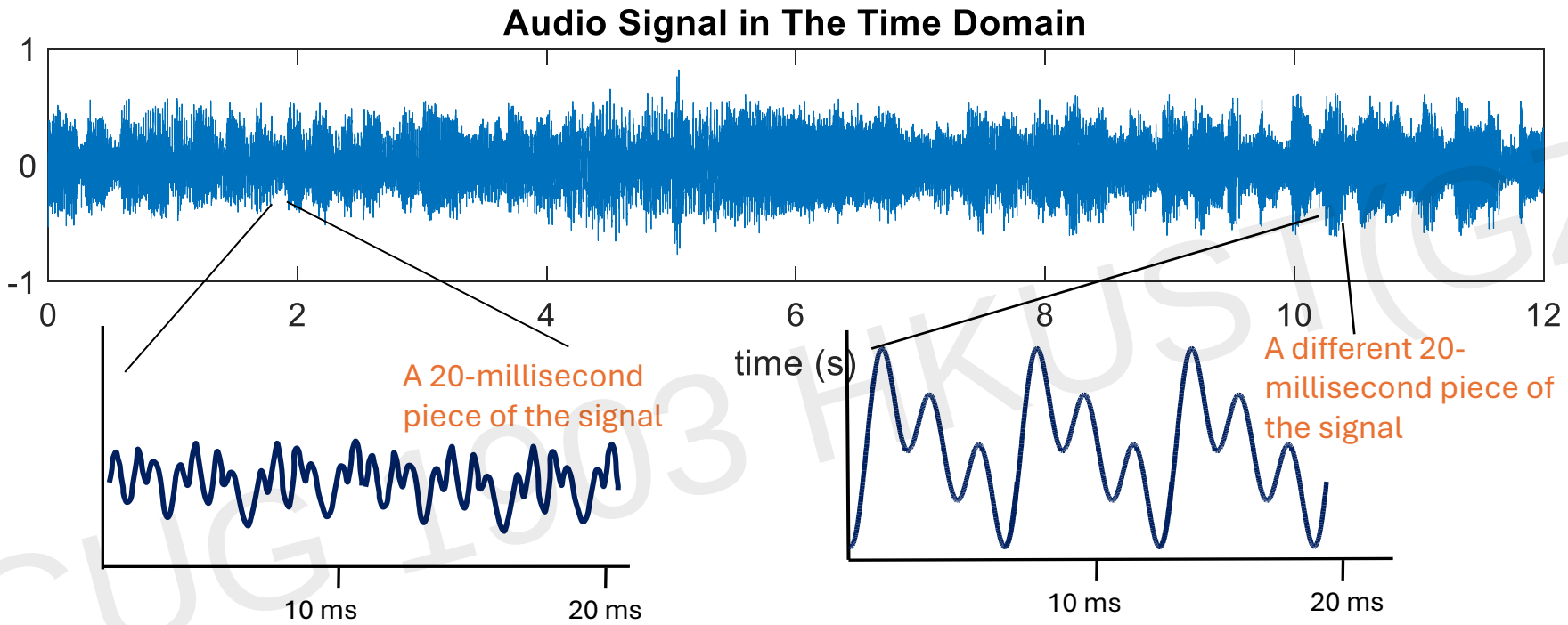
$$L_p \approx 20 \times 4.69897 \approx 93.98\ \text{dB}$$

- Is ~94 dB deafening?
  - very loud comparable to a subway train, motorcycle at close range, or a power lawn mower.
  - for continuous exposure, 94 dB can cause hearing damage over time
  - to our ears, a pressure variation of 1 Pa is not just audible, it is loud.

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Plot Sound Signal as A Function of Time

- Audio signal over 12 seconds



- When we zoom into the plot of the original sound signal to the ~20 ms time scale, we find different repetitive patterns at different parts of the music:

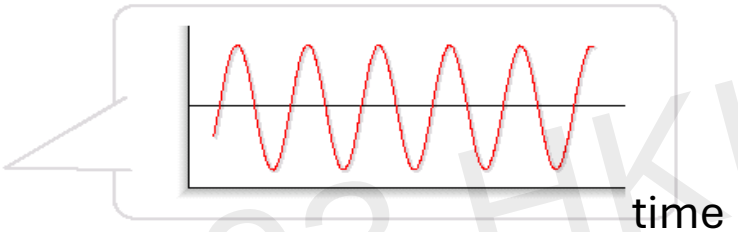
- We find that the whole piece of music is actually a sequence of *sound bites*, each lasting a fraction of a second with a recognizable repetitive patterns.

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Examples of Graphing Sound Signals

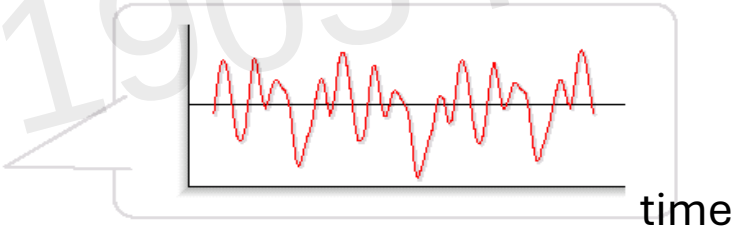
- When we plot different sound bites from different sources, we see they are all made up of *repetitive patterns*, or *oscillations*. The pattern may repeat quickly or slowly, look simple or complex, and have a small or large *amplitude*.



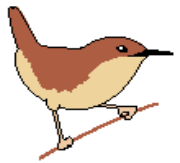


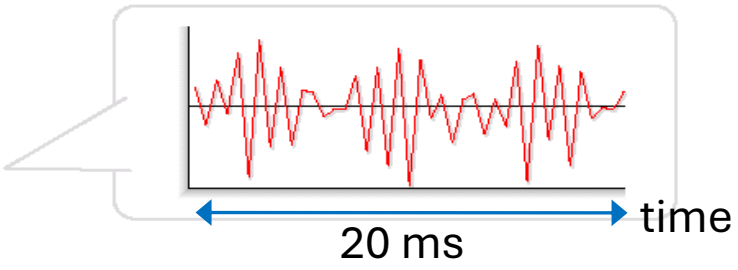
A tuning fork





A guitar





A singing bird

https://freesound.org/people

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Sound: Pitch, Loudness and Timbre

- Our ears distinguish sounds primarily by distinguishing three attributes of the sound signal:

  pitch, loudness, and *timbre*.

- The fundamental frequency of a sound signal leads to our perception of the pitch.

  The higher the fundamental frequency, the higher the pitch.

- The larger the amplitude of the oscillation, the louder is the sound.

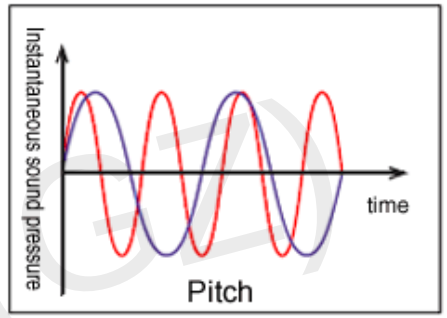
  Perception of loudness is on a logarithmic scale.

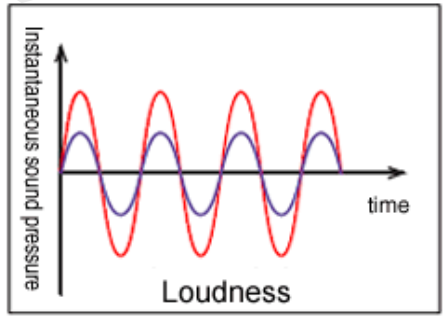
  (logarithmic scale spans wide range of values from $10^0 = 1$, $10^1 = 10$, $10^2 = 100$, ...

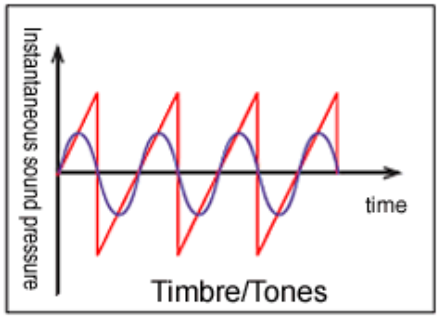
  e.g. if a person's whisper has loudness measured at 10^0,

  then a thunderstorm loudness may be measured at 10^{10}).

- *Timbre* refers to the quality of the sound. *Timbre* is determined by the details of the repetitive pattern of the sound signal.







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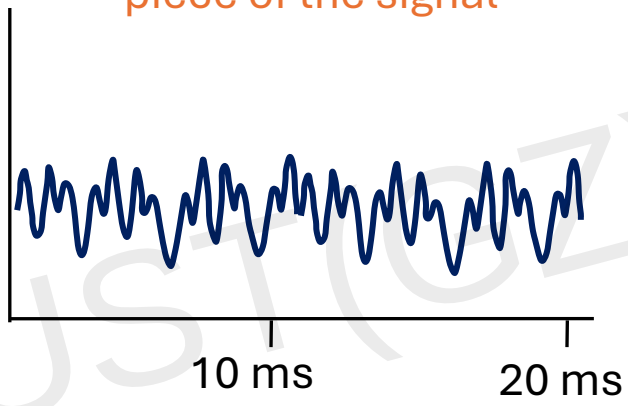
Perception of Pitch of a Sound Signal

- It is the **frequency** in the repetitive pattern in each sound bite that leads to our perception of a ***pitch***.

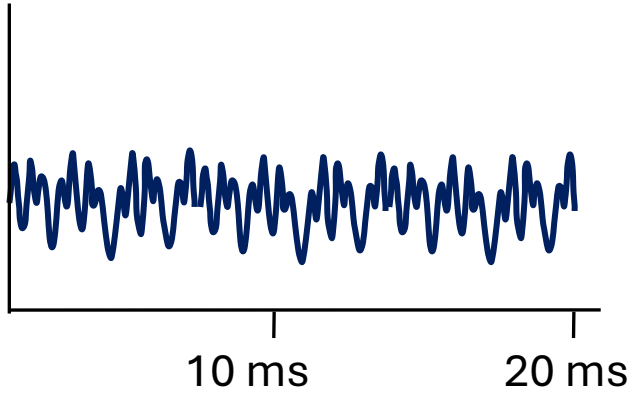
  The faster the pattern repeats, the higher the pitch.

- Speeding up or slowing down the repetition rate of the pattern leads to a different perception of the "pitch" of the sound that we hear.

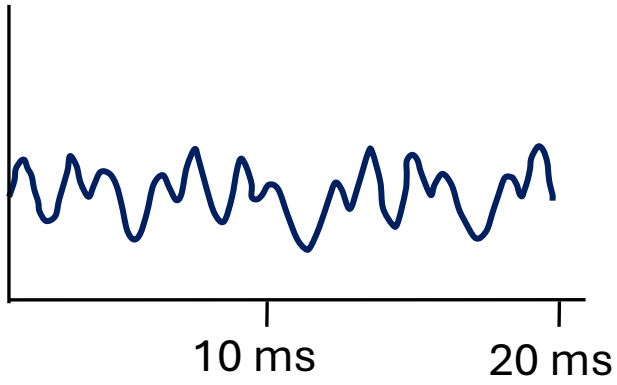
A 20-millisecond piece of the signal



Repetition rate speeded up by 1.5x



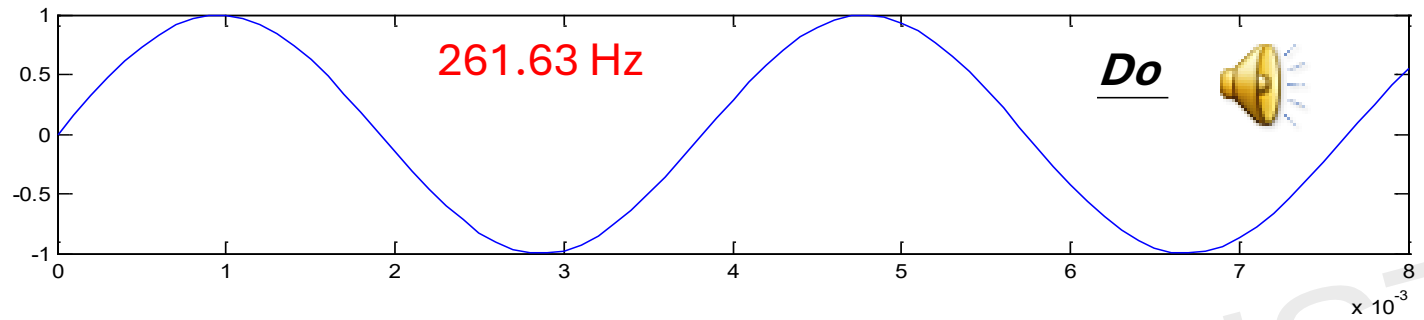
Repetition rate slowed down by 2x

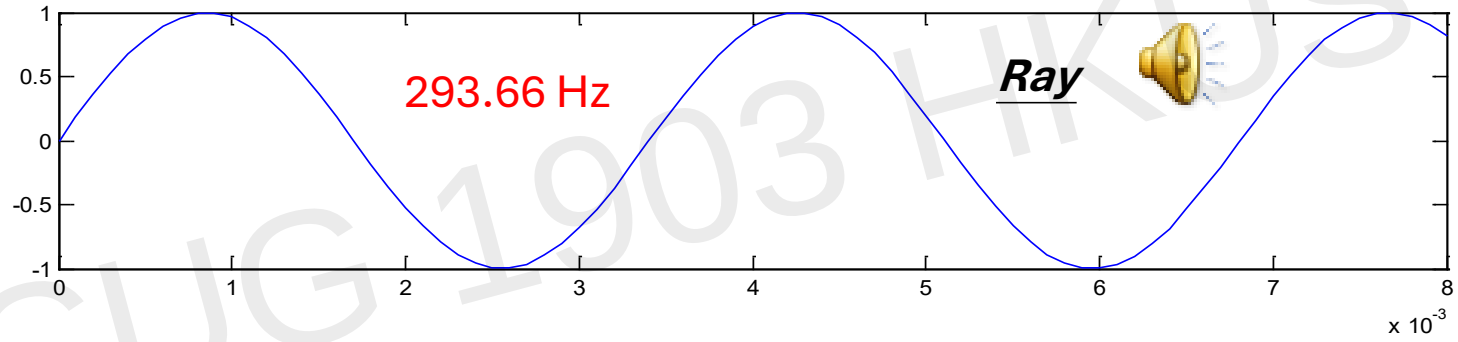


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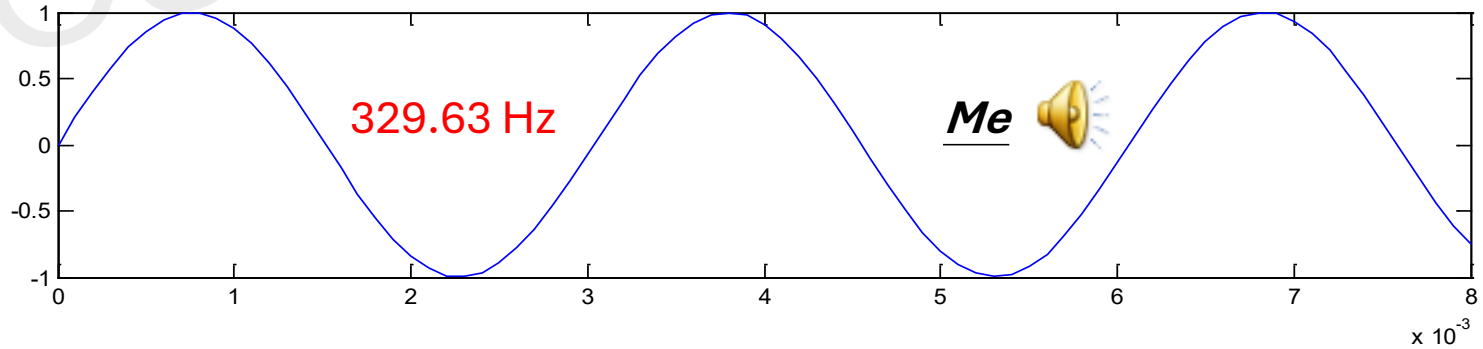
Sine Waves of Different Frequencies Give us Different Pitches

Every music note we know has a specific pitch or fundamental frequency:



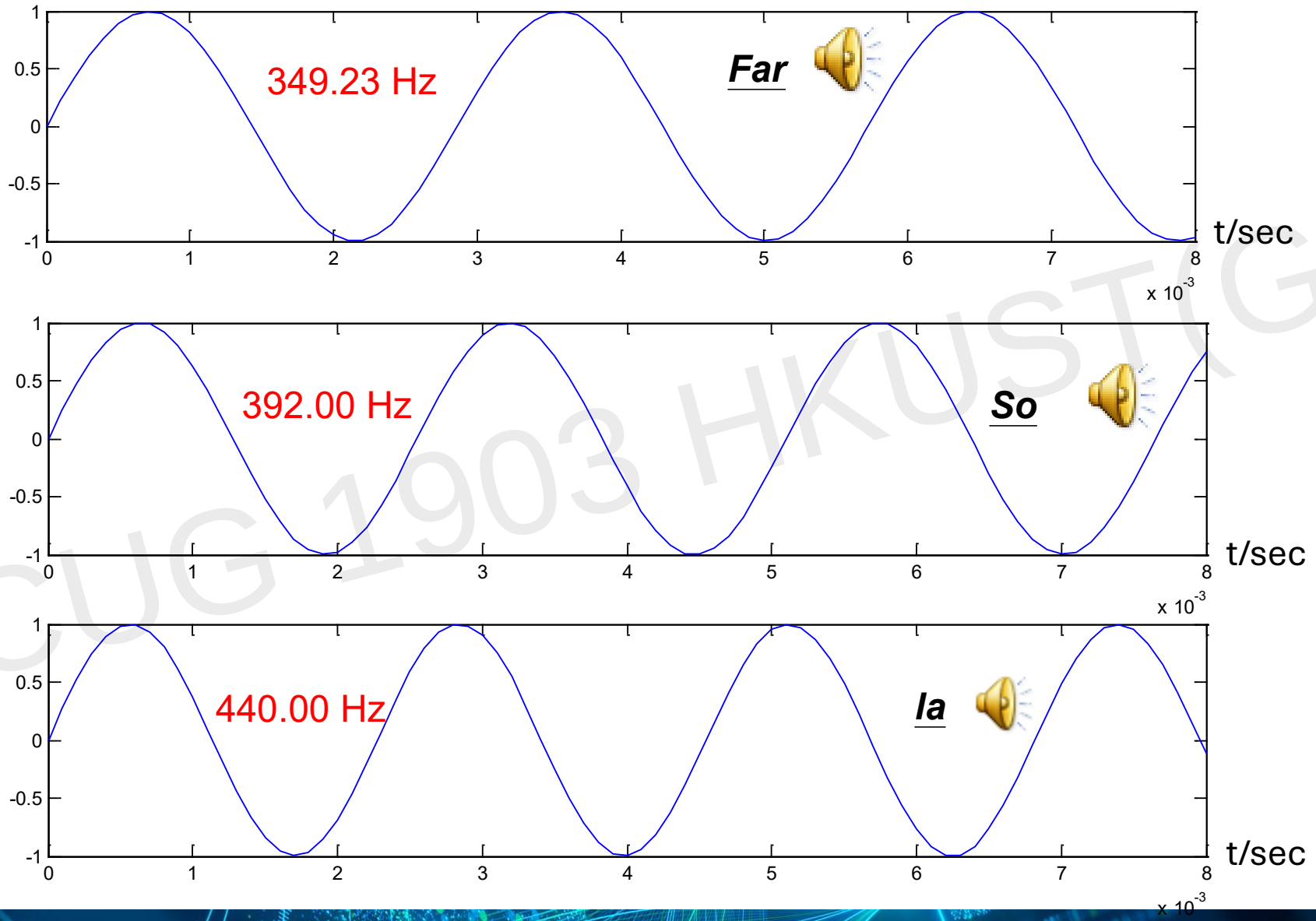


$A \sin(2\pi \cdot f \cdot t)$



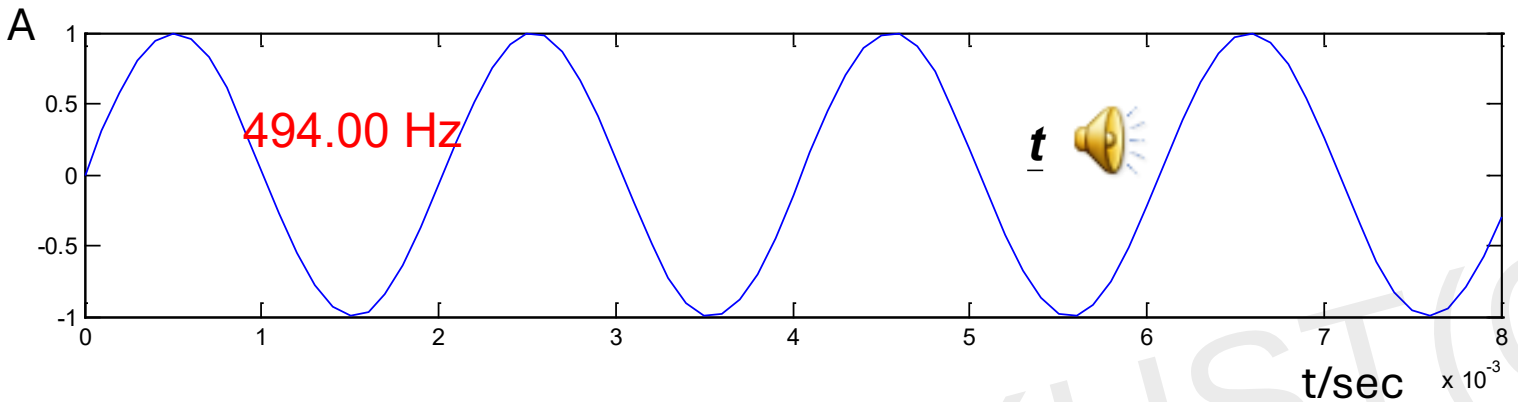
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Sine Waves of Different Frequencies Give us Different Pitches

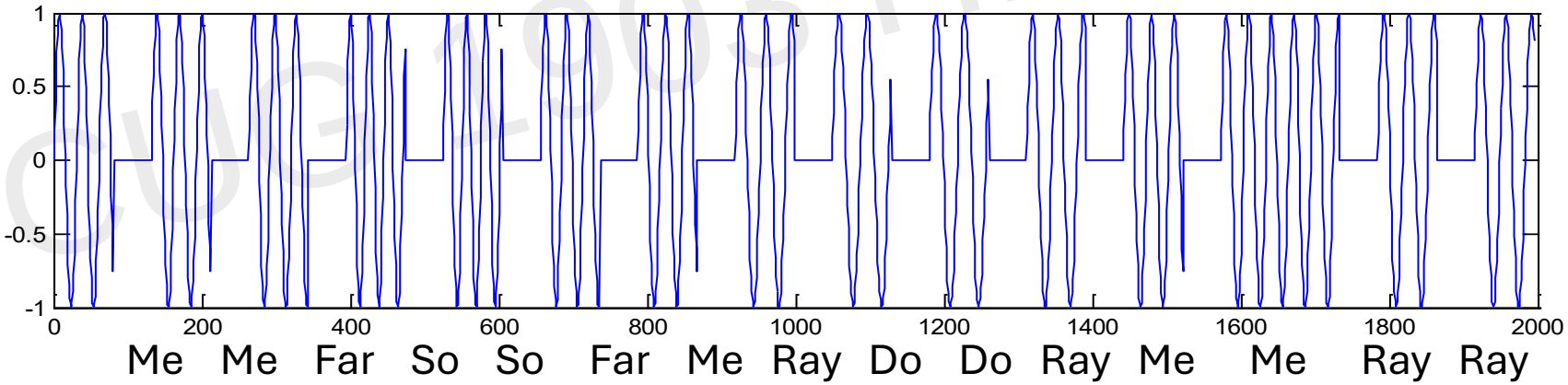


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Combining Different Notes in Time Gives Us a Song!



Combining the notes together we get a piece of music!



We have specified this piece of music as a sequence of music notes!

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Sine Waves of Different Frequencies Give us Different Pitches

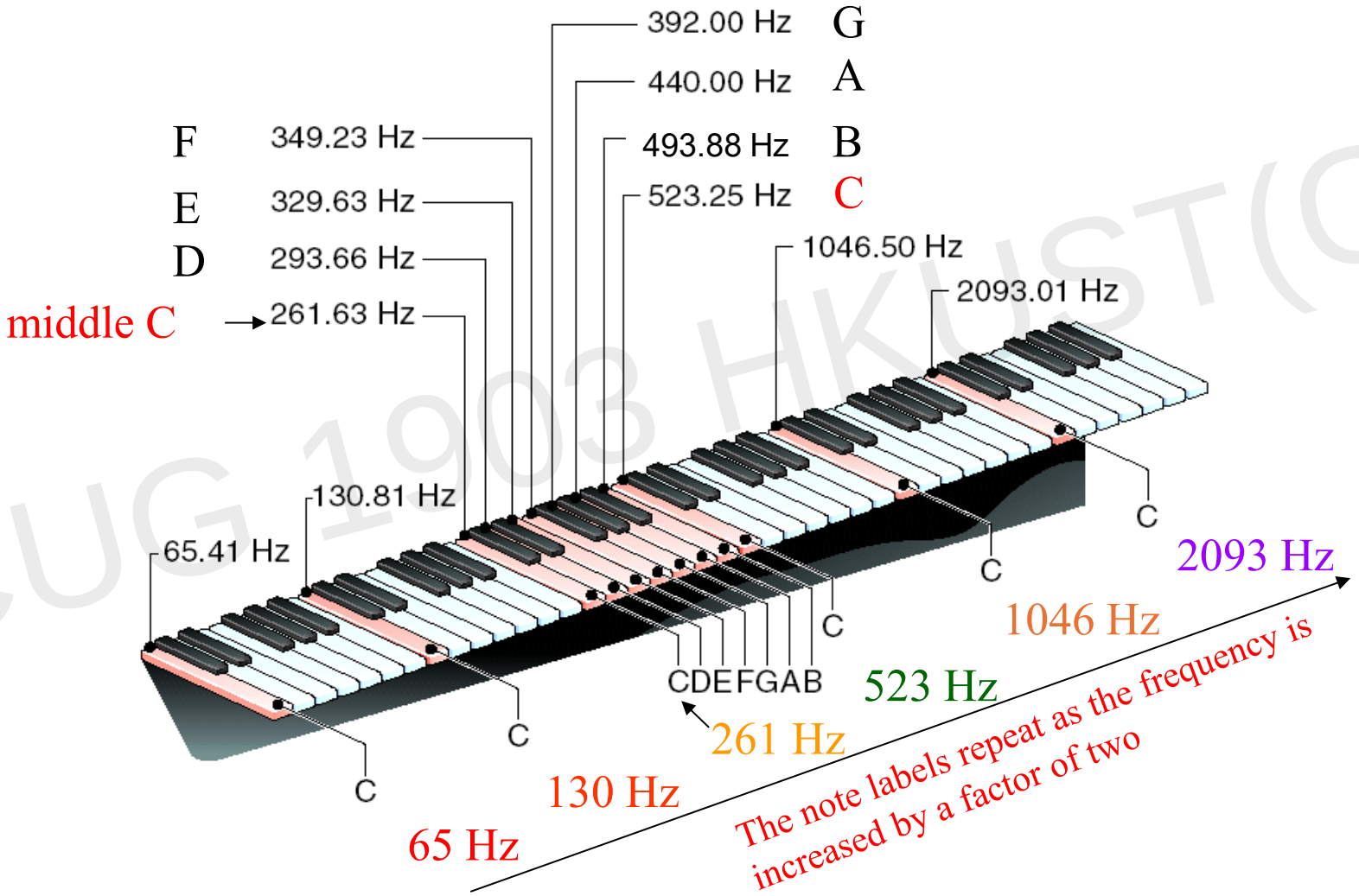
- Every music note we know has a specific pitch or fundamental frequency:

Octave	Note Name	Solfège (Fixed Do)	Approx. Frequency (Hz)
2	A2	La	110.00
2	B♭2/B2	Si♭/Si	116.54 / 123.47
3	C3	Do	130.81
3	D3	Re	146.83
3	E3	Mi	164.81
3	F3	Fa	174.61
3	G3	Sol	196.00
3	A3	La	220.00
3	B3	Si	246.94
4	**C4 (Middle C)**	**Do**	**261.63**
4	D4	Re	293.66

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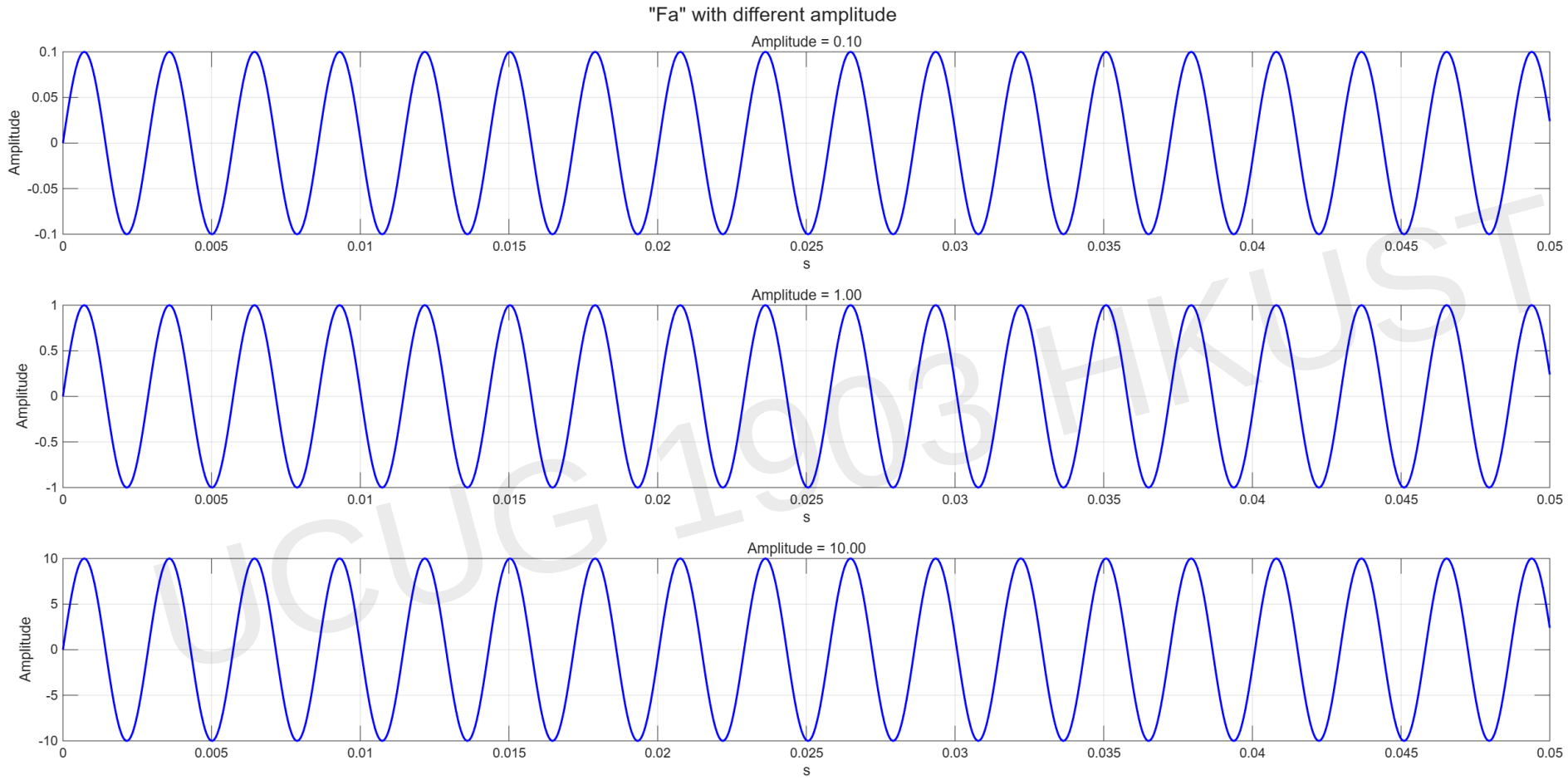
Example: Fundamental Frequencies in a Piano

- Different keys on the piano keyboard produce sounds at specific pitches, called ***notes***, labeled A to G (for the simple scale).



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Loudness: Sine Waves under Different Amplitude



Amplitude = 0.1

Amplitude = 1

Amplitude = 10

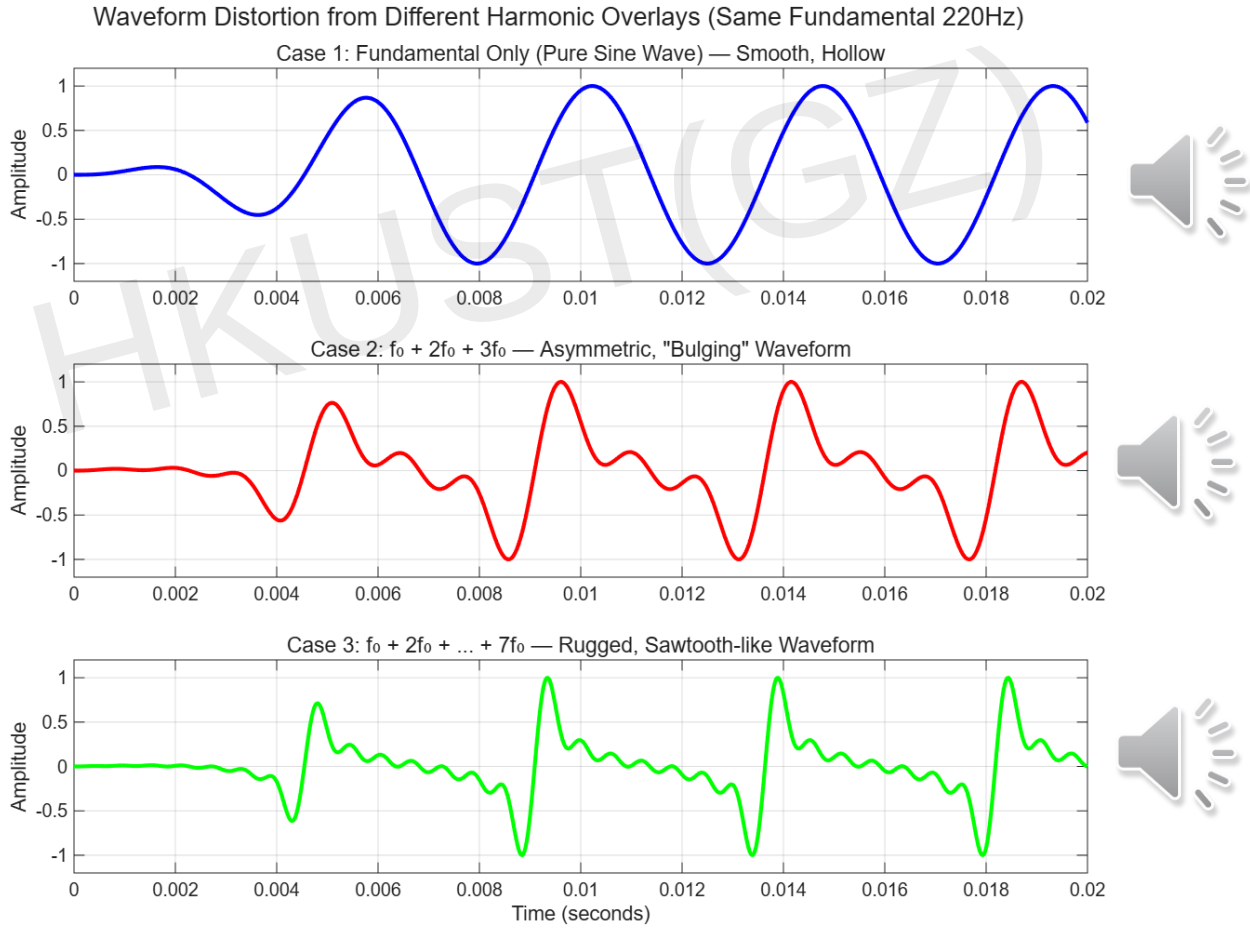
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Perception of Timbre of a Sound Signal

- A sine wave can show us pitch and loudness, but the sound it makes is nothing like what we hear in the real world.

- For example, when a piano plays the note "Do," a frequency analyzer reveals exactly the same fundamental frequency as our sinusoid wave, yet the piano's sound is much richer and more complex than the thin, sterile tone of the sine wave.

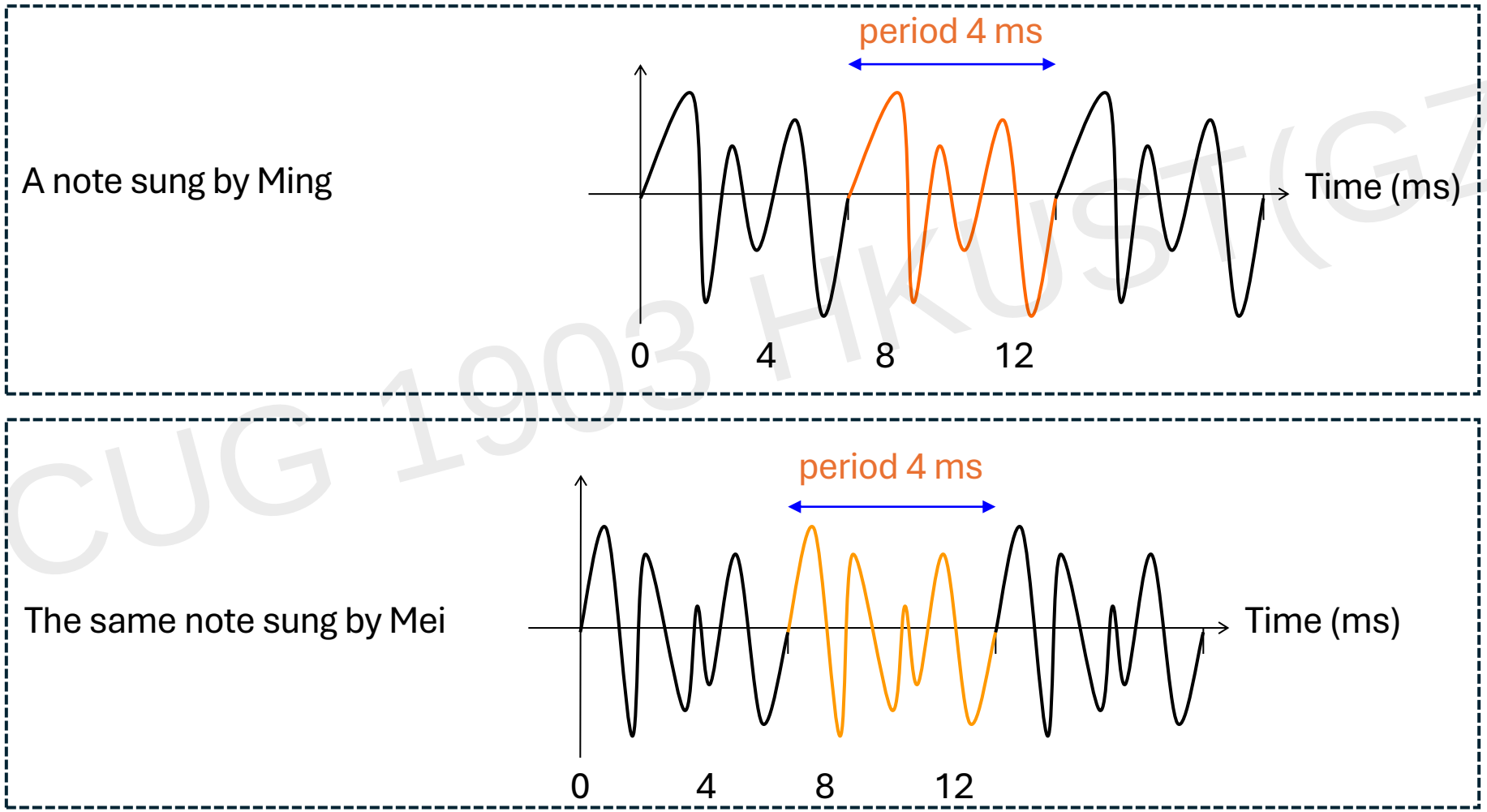
- Natural sounds consist of a fundamental frequency accompanied by its integer multiples, known as harmonics or overtones, and the specific intensity distribution among these harmonics determines the spectral richness and brightness of the sound.



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Example: Same Note Sung by Different Persons

- Different *sequences of sound bites* give us our perception of speech and music!



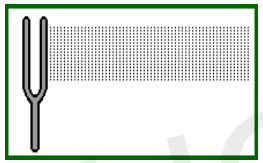
Note the different repeating structures

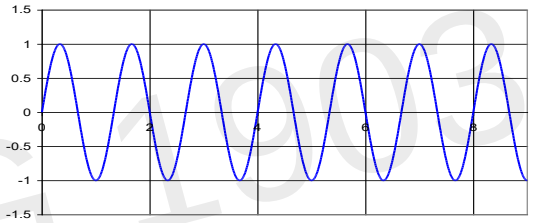
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Harmonics And “Timbre/Quality” of Sound

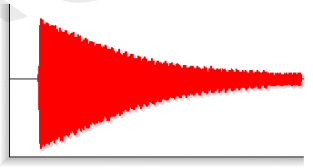
- It is the sum of the harmonics that produces the complex patterns that we observe in sound signals.
- While the pitch of the sound we hear is determined by the fundamental frequency of the repetitive pattern, the “quality” of the sound is determined by the *relative amounts (amplitudes) of the harmonics*!

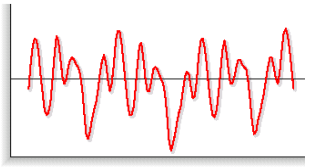
- A tone with only the *1st harmonic* (i.e. the *fundamental*) is a “*pure*” tone! Its signal is a pure sine wave (e.g. *a tuning fork*, *a nice whistle*)









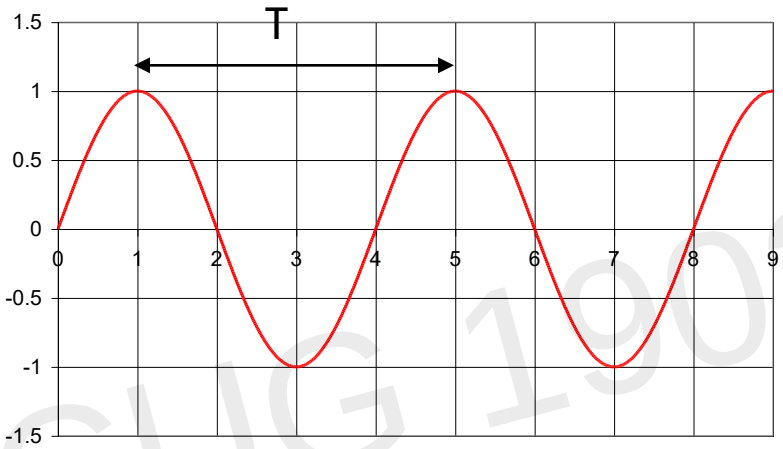


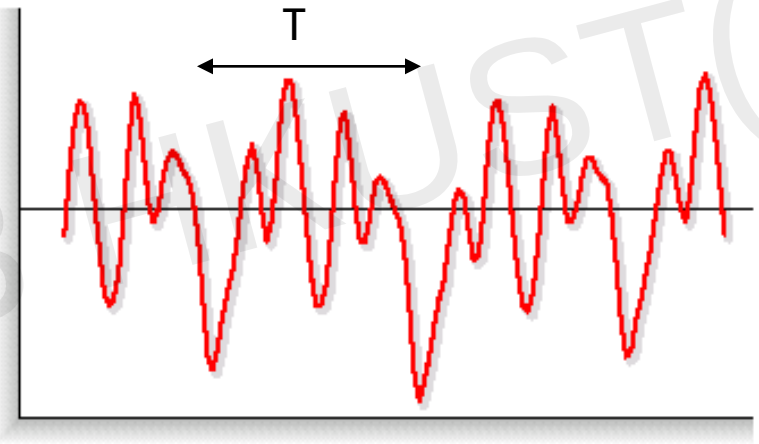
- A tone with *many strong harmonics* is a “*rich*” tone!
- So, a guitar produces a *different distribution of harmonics* than a piano or a violin or a flute or any musical instrument or you or me!

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Sound: Pitch, Loudness and Timbre

- So pure sinusoidal signals only have simple periodic variations. But periodic signals generated by piano and voices are generally more complex!





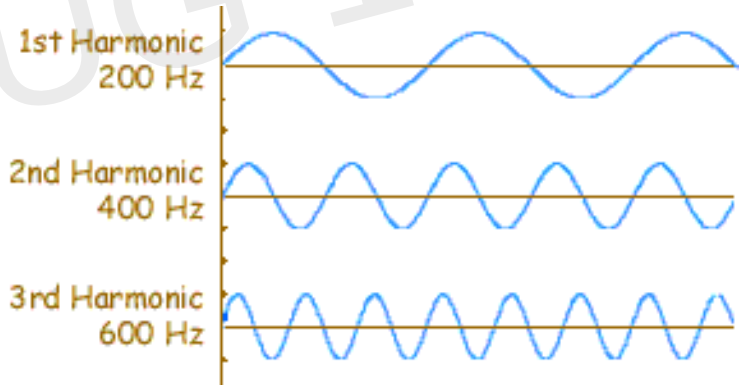
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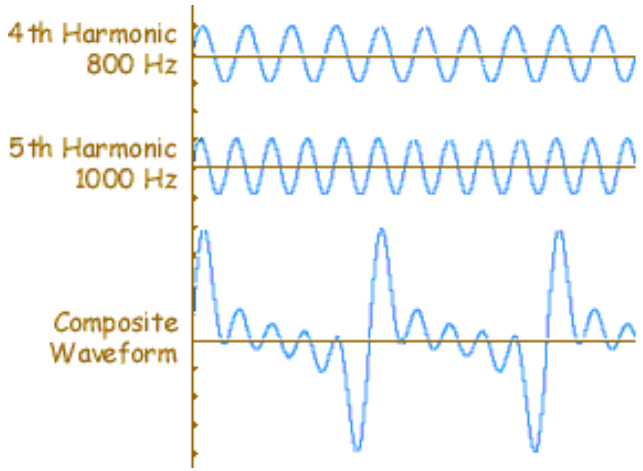
- What is in the complex pattern that leads to our perception of timbre or richness?

- ***Are there some simple ways to understand and describe the more complex periodic signals?***

Harmonics – Multiples of The Fundamental Frequency

- We can answer our question by introducing the concept of ***harmonics***. It is the harmonics that produce different repetitive patterns of the *same* period and give us the perception of timbre or richness in sound.
- Harmonics are sine waves at ***n (an integer) times the fundamental frequency*** (n = 1, 2, 3, ...)
  - ***Adding harmonics together does not affect the fundamental frequency*** f and period $T = 1/f$, because all harmonics repeat after the same time period T.
  - Periodic signals with different repetitive patterns contains different amount/amplitudes of harmonics.





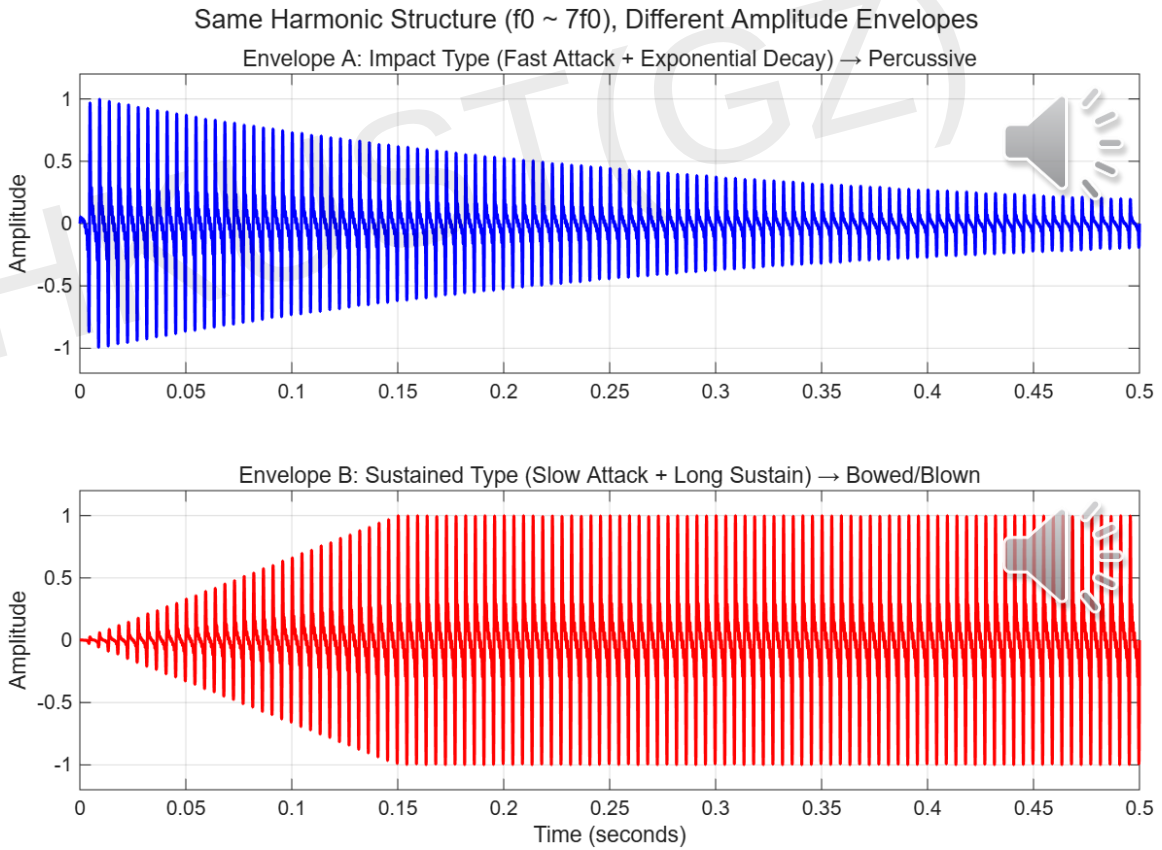
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Perception of Timbre of a Sound Signal

- Different combinations of harmonics (spectral structure) give us different tone colors, from pure tones to rich complex tones.

- However, **harmonics alone are insufficient to account for the full variety of sounds we hear daily** – because in reality, almost no sound maintains a constant loudness over time.

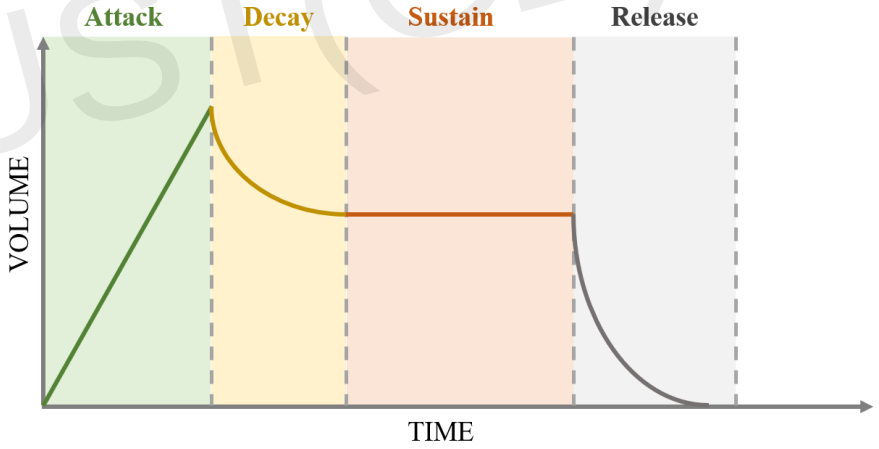
- This leads us to another crucial factor – the **temporal envelope**.



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Envelope

- While the sine waves consists sound signals, the loudness of these components is rarely constant over time.
- The curve that describes how the overall amplitude (loudness) of a sound changes over time is known as its **amplitude envelope**.
- A standard model for describing the envelope is the **ADSR** model:
  - **A**ttack: Time taken to rise from zero to peak amplitude.
  - **D**ecay: Time taken to fall from peak to the sustain level.
  - **S**ustain: The amplitude level maintained while the note is held.
  - **R**elease: Time taken to fade from sustain to zero after the note ends.

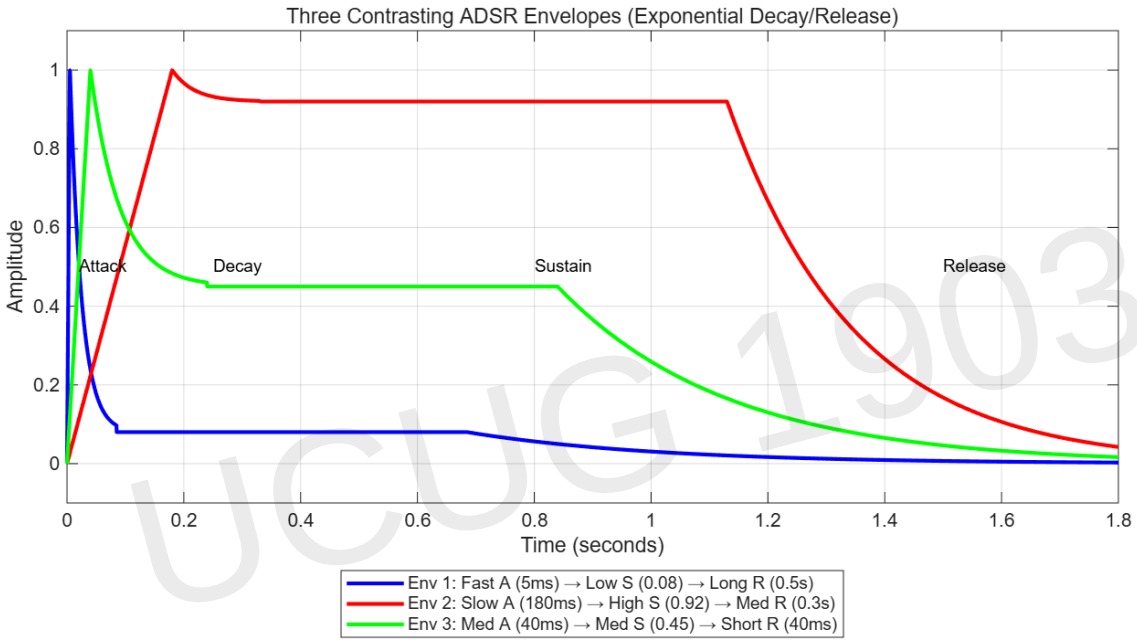


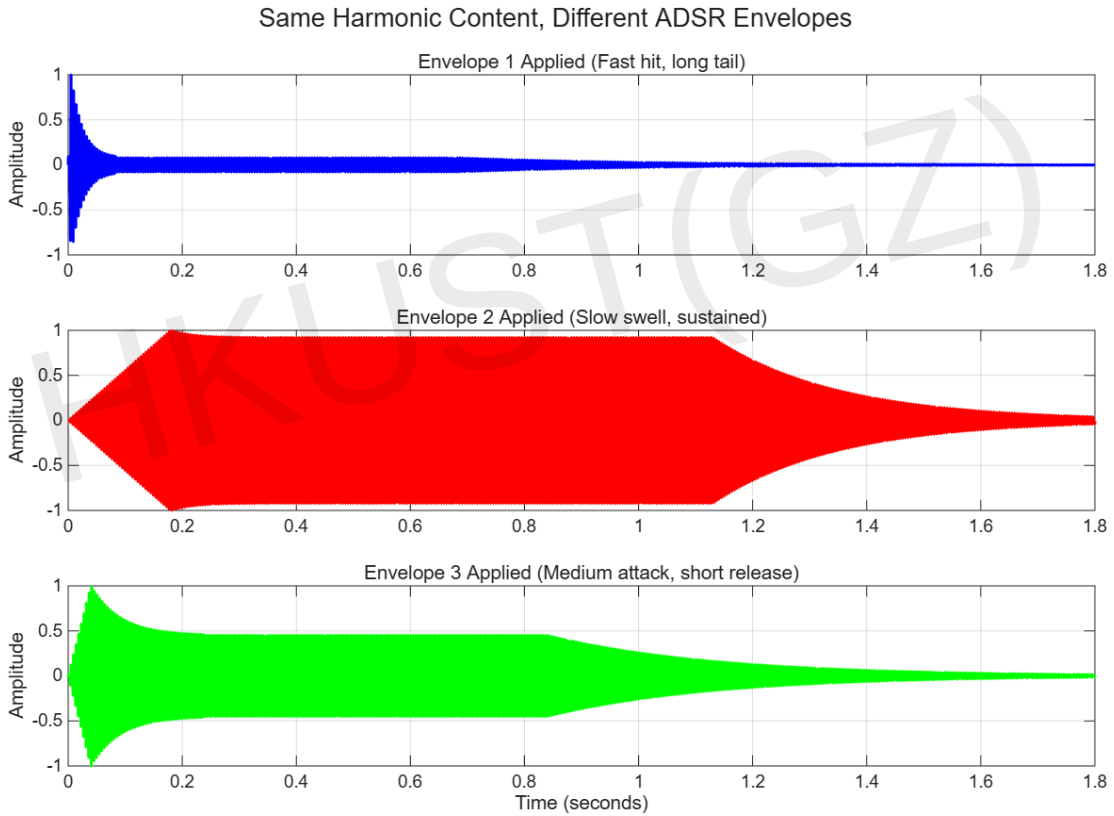
- Applying an envelope is mathematically equivalent to **multiplying** it with the waveform

$$y(t)=x(t)\cdot e(t)$$

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Envelope Examples



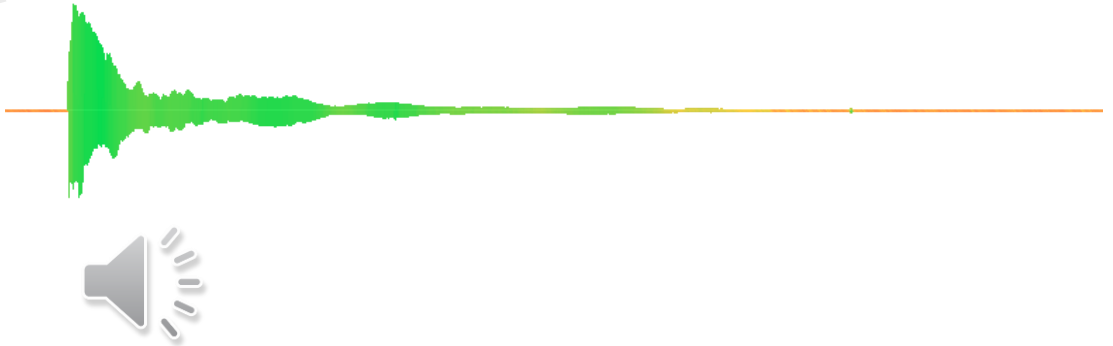


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Timbre and Envelope of a Sound Signal

- Two Physical Determinants
  - The distribution of harmonics (overtones) determines the spectral richness.
  - The amplitude envelope (ADSR: Attack, Decay, Sustain, Release) determines how the sound evolves over time.





https://freesound.org/people/ldk1609/sounds/56204/

https://freesound.org/people/nanliu_music/sounds/847227/

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Summary

- Sound is the variation in acoustic pressure that reaches our ears.
- Periodic signals are highly related to sound signals and serve as their mathematical foundation.
- Three Elements of a Sound Signal: Pitch, Loudness, Timbre
  - The *pitch* of sound is determined by the ***fundamental frequency*** of the sound signal.
  - The *loudness* of sound is determined by the ***envelope amplitude (intensity)*** of the signal
  - The *timbre*, of sound is determined by the amount of different ***harmonics***. (We will discuss more on harmonics in Lecture 04.)
- The human auditory system primarily responses to the different ***sine waves*** in the sound signal.

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https://www.youtube.com/watch?v=5tGEDgkZlC8

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Fundamental Frequency and Harmonics

- We introduced the fact that our ear basically responses to the ***sine waves*** in sound signals.

For *any* sinusoidal signal with frequency *f*,

sinusoidal signals with frequencies of "**integer** × ***f***" are known as its *harmonics*.

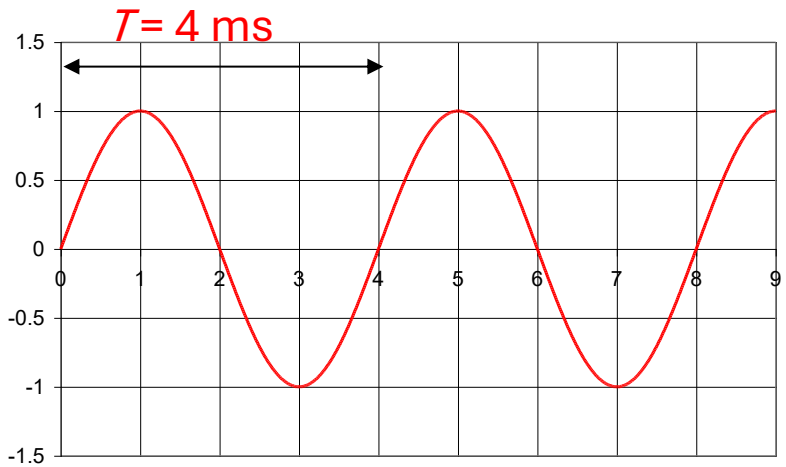
(e.g. *f* = 250 Hz, 2*f* = 500 Hz, 3*f* = 750 Hz, 4*f* = 1000 Hz)

(e.g. 375 Hz = 1.5f is not a harmonic of f)

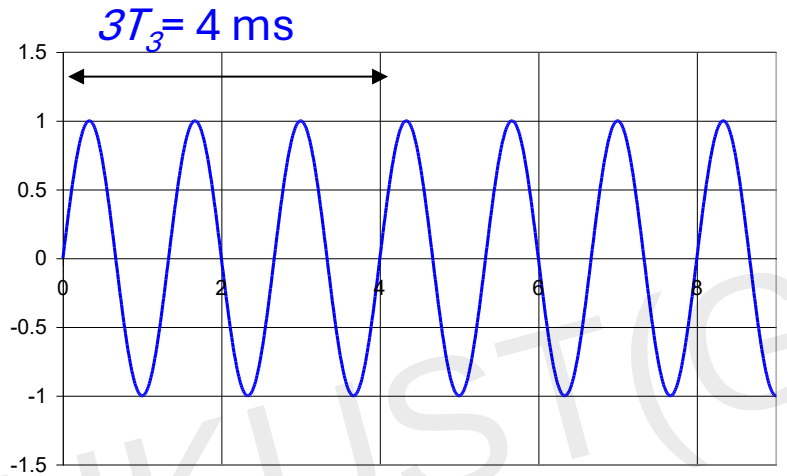
- Sinusoidal signal with frequency "*f*" is the "1st harmonic." ("*f*" is also known as the "fundamental frequency.")
- Sinusoidal signal with frequency "2*f*" is the "2nd harmonic."
- Sinusoidal signal with frequency "3*f*" is the "3rd harmonic," and so forth.

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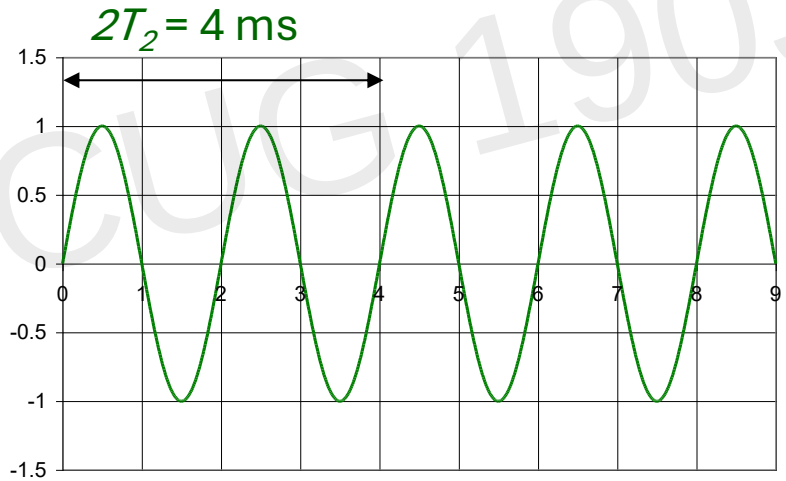
Examples of Harmonics



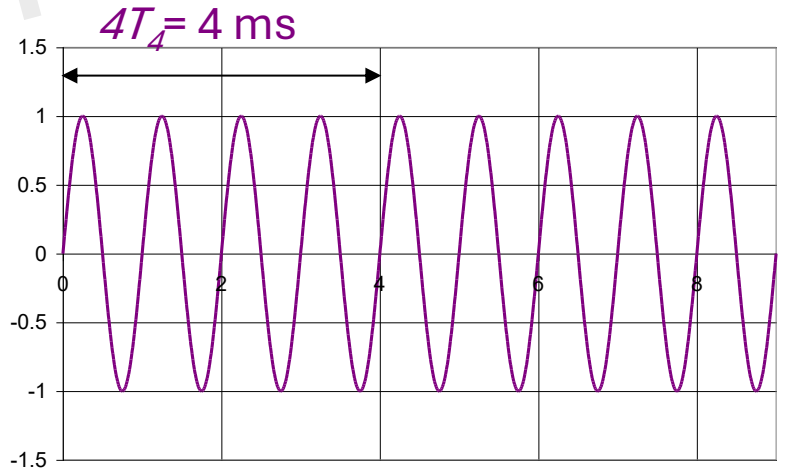
Fundamental = f = 1/4ms = 250 Hz



3rd harmonic = 3f = 3 × 1/4ms = 750 Hz



2nd harmonic = 2f = 2 × 1/4ms = 500 Hz



4th harmonic = 4f = 4 × 1/4ms = 1000 Hz

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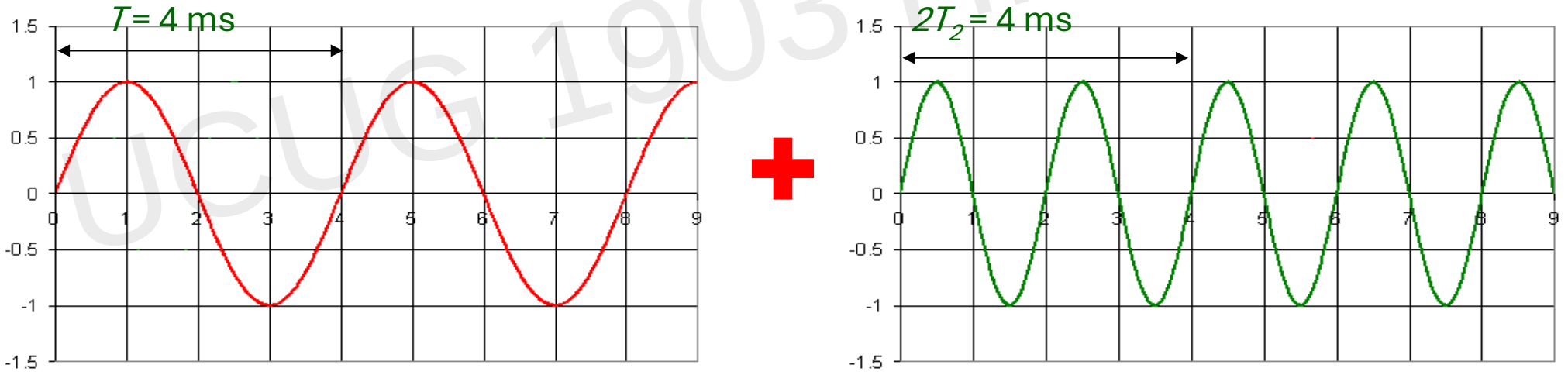
Example – Making Complex Periodic Signals

- Adding harmonics does not change the fundamental frequency (period) of a signal,

  i.e. the signal still repeats with period T

- In the example below, we add together a sine wave at 250 Hz with its 2nd harmonic (sine wave at 500 Hz) of equal amplitude:

Example:



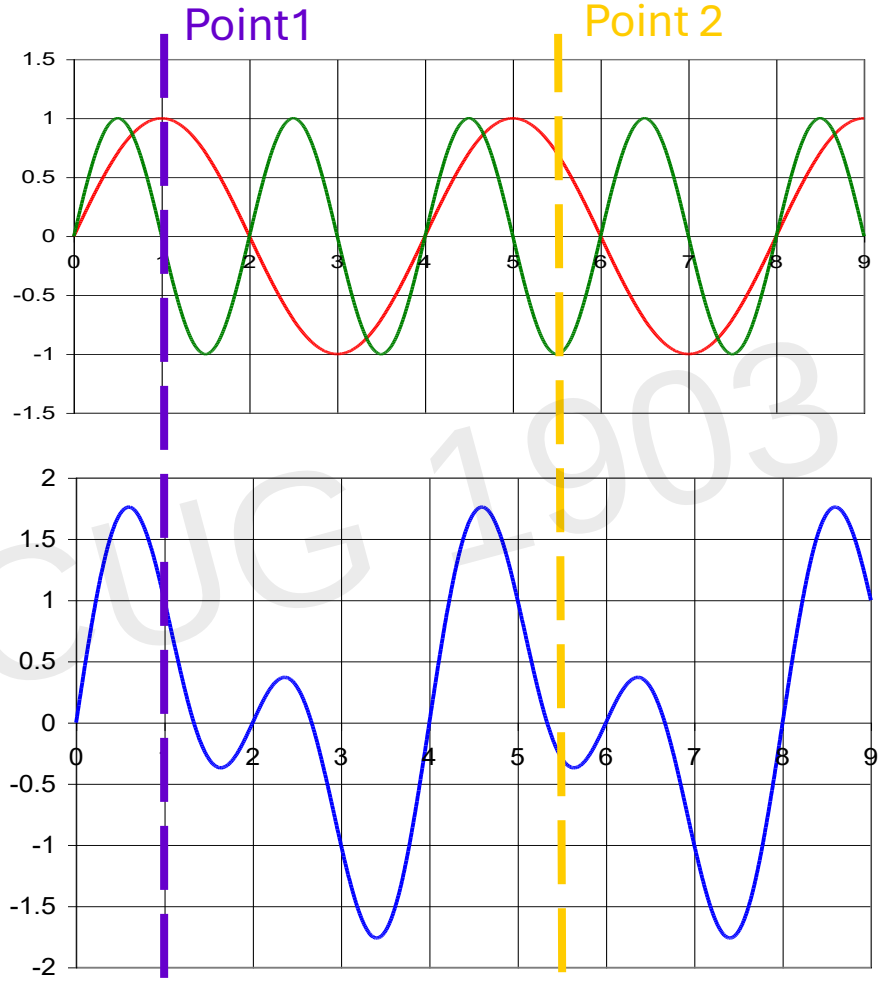
Fundamental = f = 1/4ms = 250 Hz

2nd harmonic = 2f = 2 × 1/4ms = 500 Hz

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Example – Making Complex Periodic Signals

- At each point, we add the value of the two sine waves. The result is a more complex waveform. But the fundamental frequency (period **T**) remains the same.



– At Point 1,

Red Signal = 1

Green Signal = 0

Resultant Signal = 1 + 0 = 1

– At Point 2,

Red Signal = 0.7

Green Signal = -1

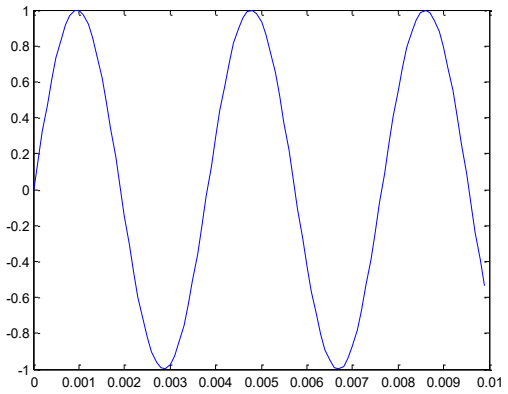
Resultant Signal = 0.7 + (-1) = -0.3

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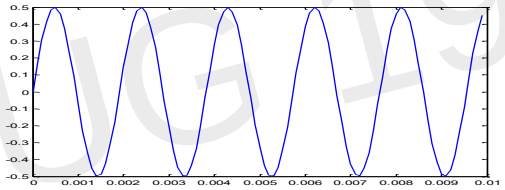
Example – Adding Different Amounts of Harmonics

- We can add different amounts of harmonics. ***The fundamental frequency remains the same.***

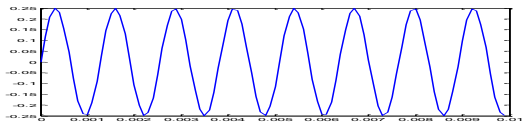
1st harmonic (Fundamental frequency)

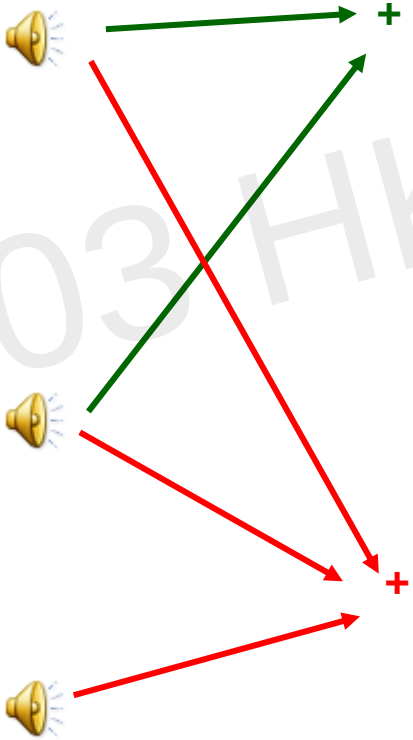


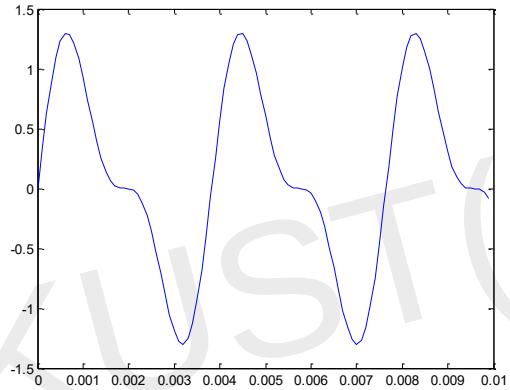
2nd harmonic



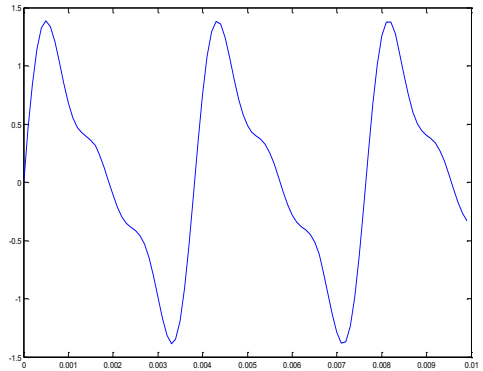
3rd harmonic







1st harmonic + 2nd harmonic



1st harmonic + 2nd harmonic+ 3rd harmonic

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Proof of Proposition

- We construct a periodic signal by adding sine waves at these harmonic frequencies with arbitrary amplitudes A_n and ϕ_n:

$$s(t) = \sum_{n=1}^{N} A_n \sin(2\pi n f_0 t + \phi_n)$$

The n-th harmonic is:

$$s_n(t) = A_n \sin(2\pi n f_0 t + \phi_n)$$

Its period is:

$$T_n = \frac{1}{n f_0} = \frac{T}{n}, \quad \text{where } T = \frac{1}{f_0}.$$

So $s_n(t)$ repeats every T_n, and since T_n divides T exactly (because $T/T_n = n$ is an integer), $s_n(t)$ also repeats every T.

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Proof of Proposition

- Summing all signals with common period T

If each $s_n(t)$ satisfies $s_n(t+T) = s_n(t)$ for all t, then their sum satisfies:

$$s(t+T) = \sum_{n=1}^{N} s_n(t+T) = \sum_{n=1}^{N} s_n(t) = s(t).$$

Thus, $s(t)$ is periodic with period T.

- The smallest period is T if $A_1 \neq 0$

If T' were a period, then $s(t+T') = s(t)$ for all t.
Consider the first harmonic $s_1(t) = A_1 \sin(2\pi f_0 t + \phi_1)$.
For $s_1(t+T') = s_1(t)$ to hold for all t, we need:

$$\sin(2\pi f_0(t+T') + \phi_1) = \sin(2\pi f_0 t + \phi_1) \quad \forall t.$$

This implies $2\pi f_0 T' = 2\pi m$ for some integer m, so $T' = m/f_0 = mT$.

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Proof of Proposition

- The smallest feasible positive $T^{'}$ is T

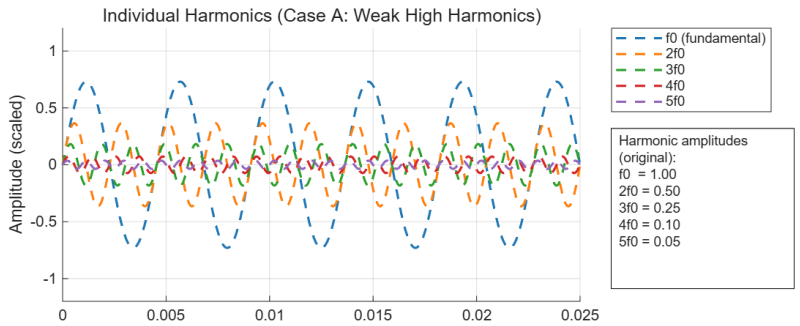
The **smallest positive** T' satisfying this for $m = 1$ is T.
If $m > 1$, then T' would be larger than T.
Thus the **minimal period** of $s_1(t)$ alone is T.

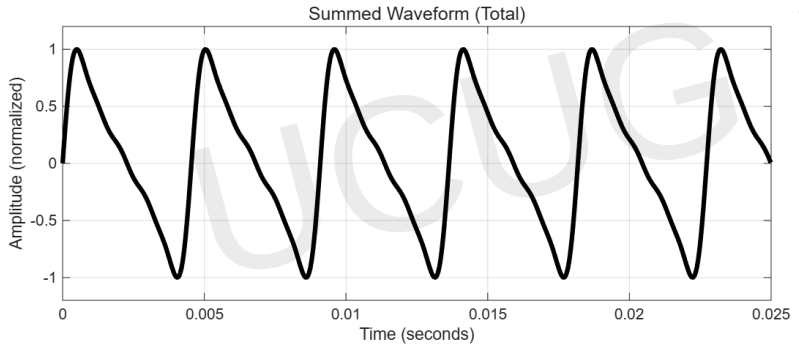
- Therefore, the fundamental period is T, and the fundamental frequency is $f_0 = 1/T$.

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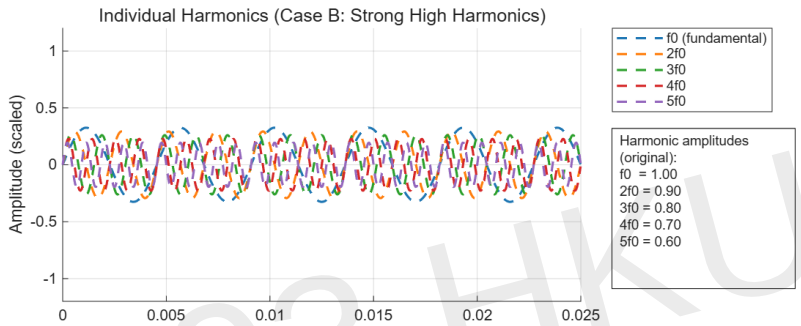
Harmonic with Different Amplitudes

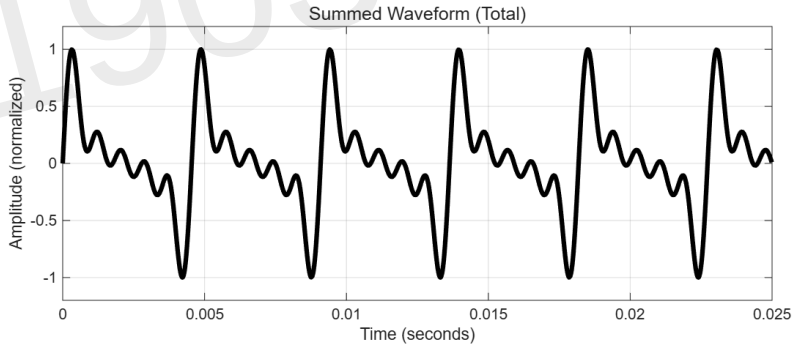
- In real-world sounds, the **relative amplitudes** of the harmonics (the Spectral Envelope) carry the **identity information** of the sound source.



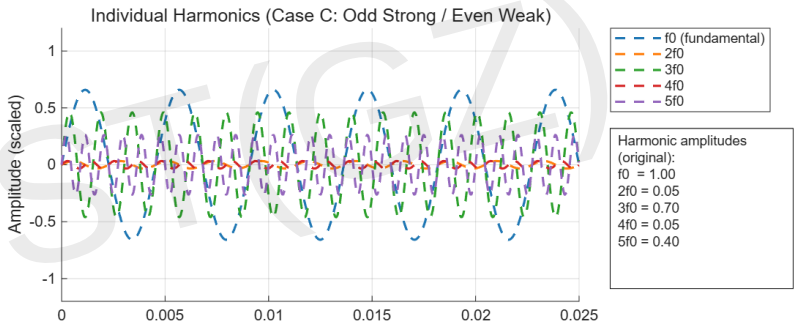


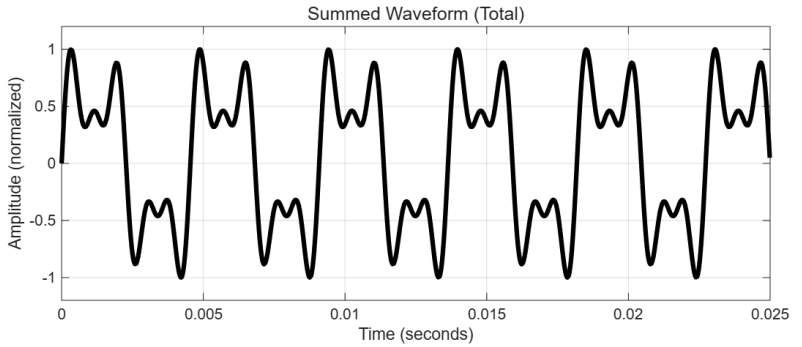














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Fourier Series

- A ***Fourier series*** is an infinite series of the form

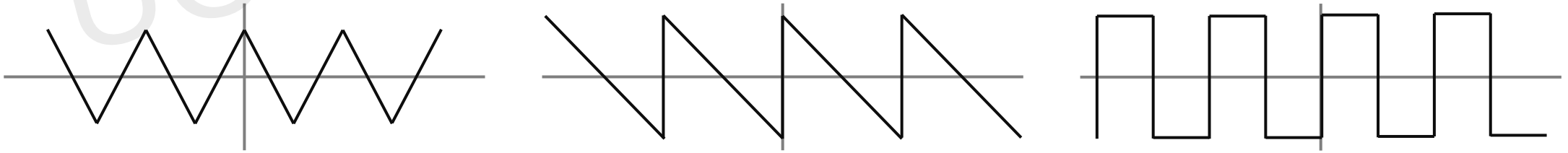
$$a + \sum_{n=1}^{\infty} \mathrm{b}_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} c_n \sin(n\omega_0 t)$$

The sine and cosine terms oscillating at the frequency $n\omega_0$ correspond to the n-th harmonic



Joseph Fourier
French mathematician
(1768-1830)

- Virtually *any periodic function* that arises in applications can be represented as the sum of a Fourier series.



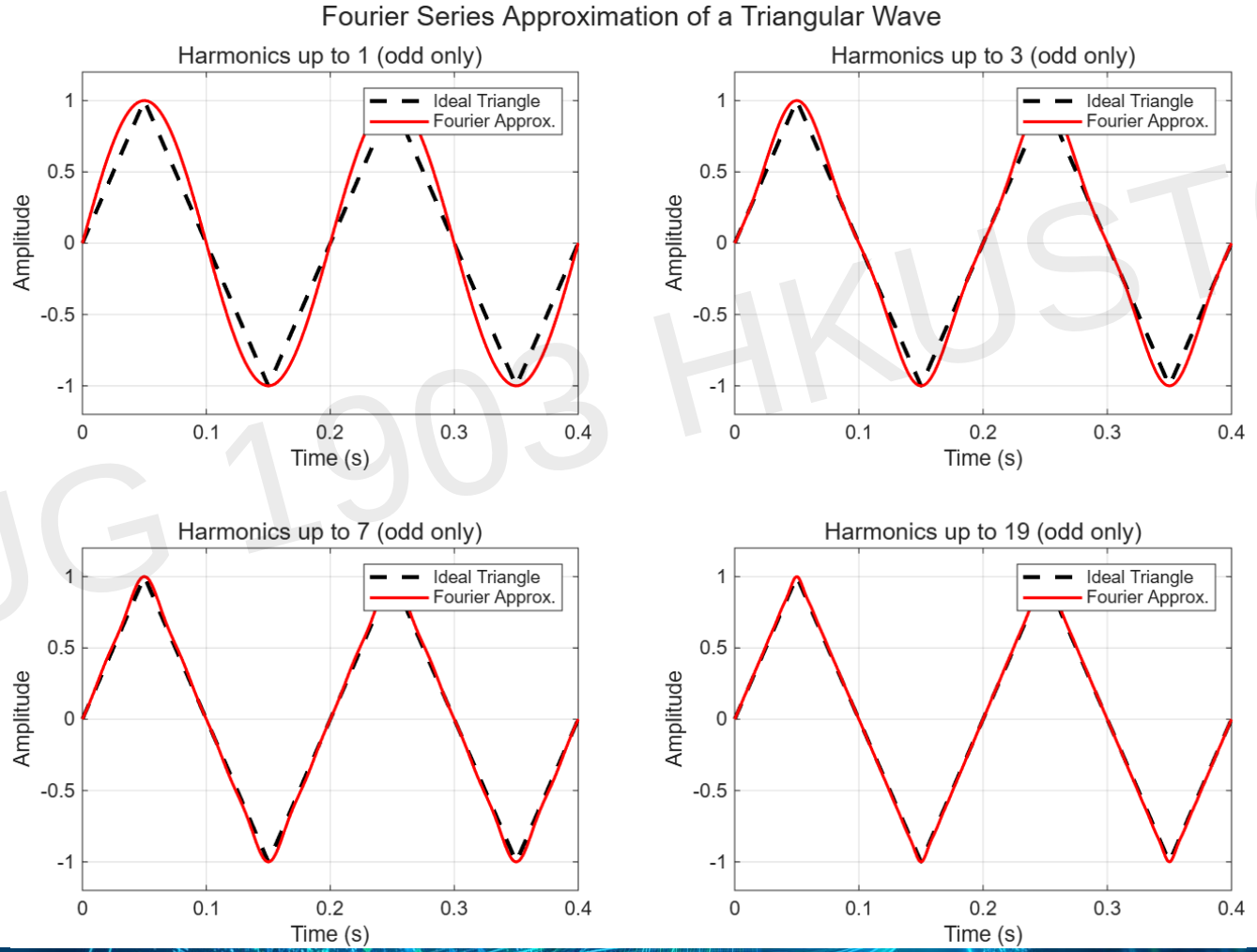
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https://www.youtube.com/watch?v=DFRBL2FsTC4

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Fourier Series

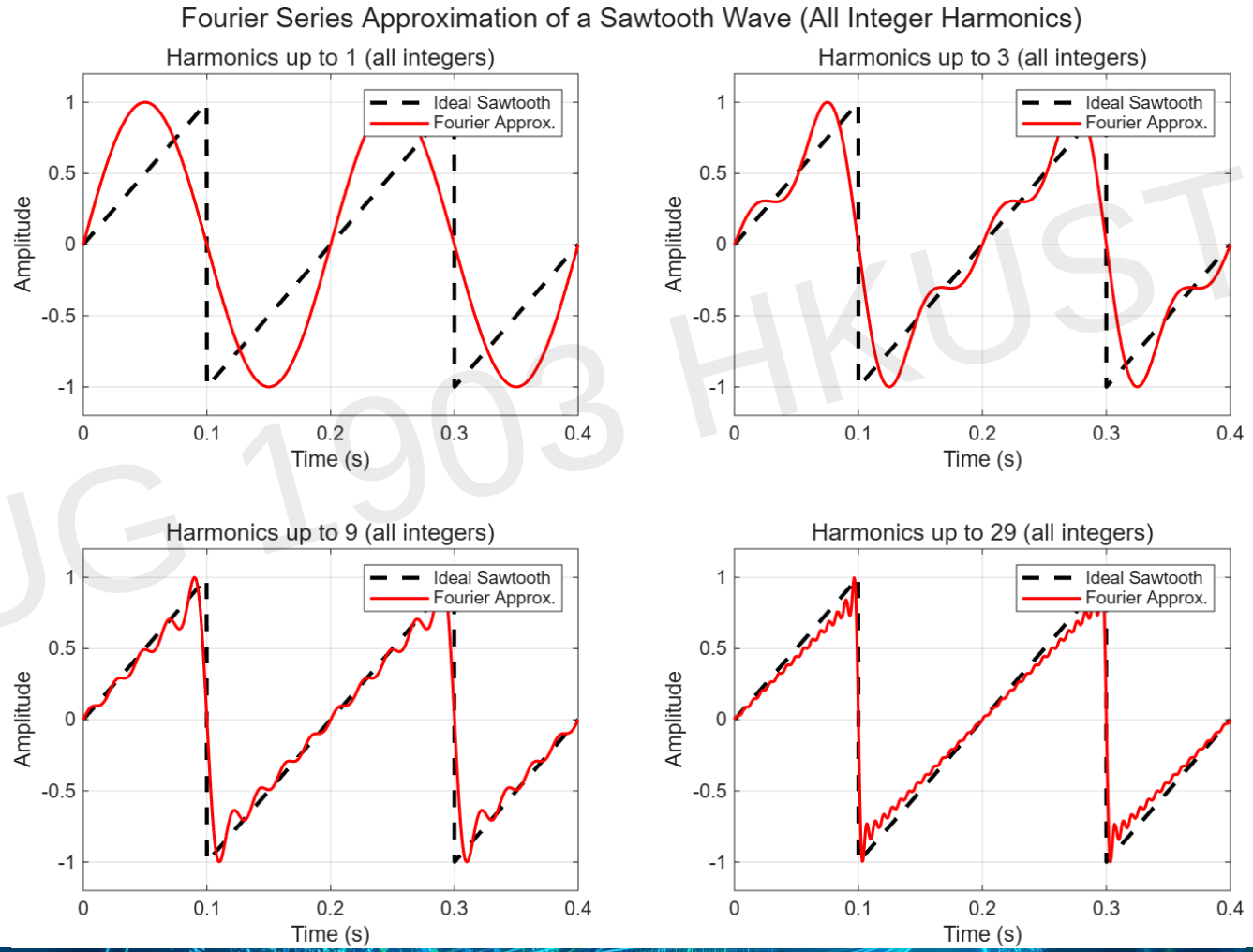
- Triangle Wave : $\Lambda(t) = \cos t + \frac{\cos 3t}{3^2} + \frac{\cos 5t}{5^2} + \frac{\cos 7t}{7^2} + \frac{\cos 9t}{9^2} + \cdots$



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Fourier Series

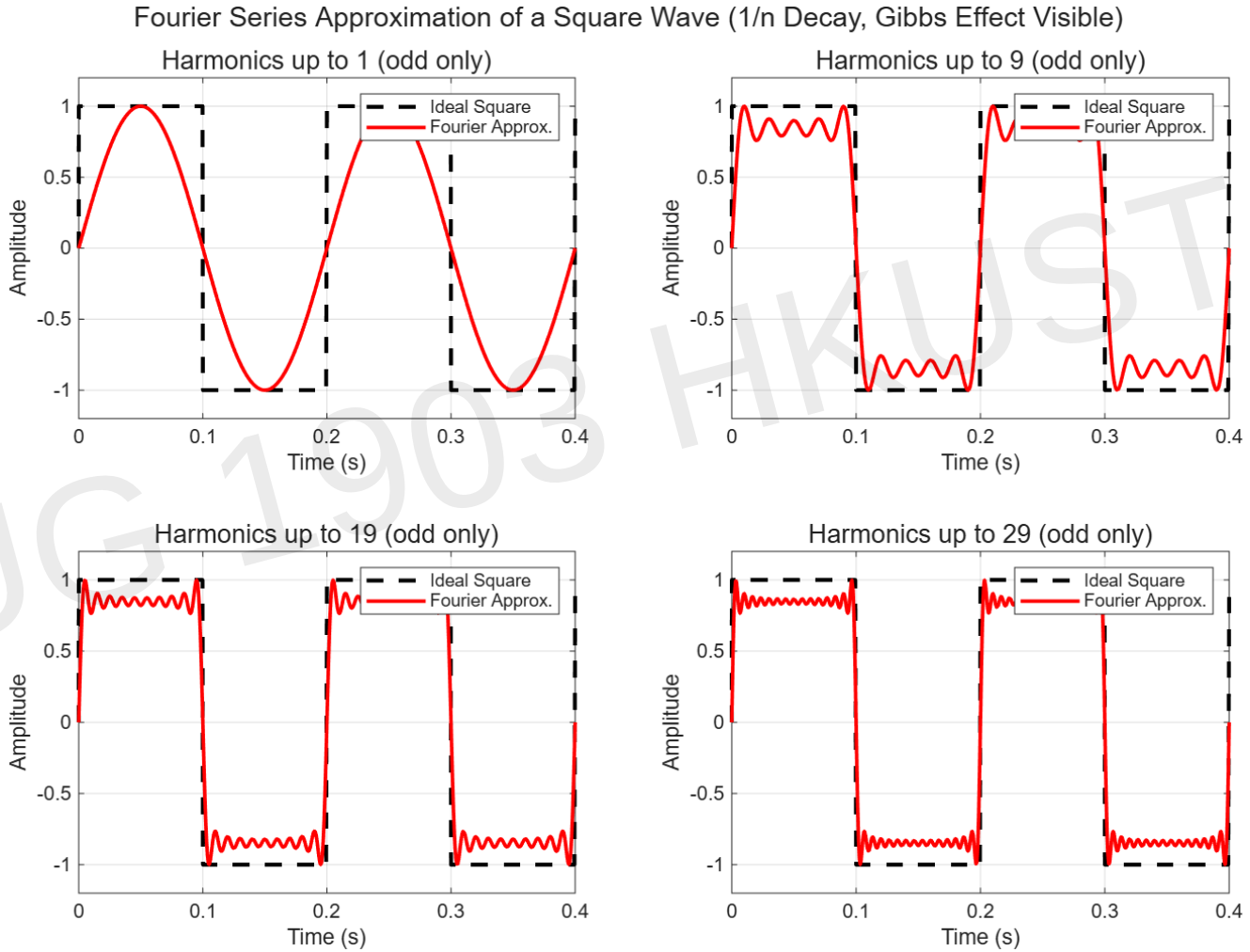
- Sawtooth Wave: $N(t) = \sin t + \frac{\sin 2t}{2} + \frac{\sin 3t}{3} + \frac{\sin 4t}{4} + \frac{\sin 5t}{5} + \cdots$



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Fourier Series

- Sawtooth Wave: $\sqcap(t) = \sin t + \frac{\sin 3t}{3} + \frac{\sin 5t}{5} + \frac{\sin 7t}{7} + \frac{\sin 9t}{9} + \cdots$



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Trigonometric Polynomial

- A trigonometric polynomial is a polynomial expression involving $\cos x$ and $\sin x$

$$\cos^5 x + 6\cos^3 x \sin^2 x + 3\sin^4 x + 2\cos^2 x + 5 \quad \Rightarrow \quad 5\cos^3 x \sin^2 x + 3\sin^4 x + \cos^3 x - 2\sin^2 x + 7$$

- Why does the above hold? $\cos^2 x + \sin^2 x = 1$

Most trigonometric polynomials can be written in several different ways → Non-unique form

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Fourier Sum

- A Fourier sum is a Fourier series with finitely many terms:

$$5 + 3\sin 2x + 4\cos 5x - 3\sin 5x + 2\cos 8x.$$

- Every Fourier sum is actually a trigonometric polynomial, and any trigonometric polynomial can be expressed as a Fourier sum

$$\cos A \cos B = \frac{1}{2}\cos(A-B) + \frac{1}{2}\cos(A+B)$$

$$\sin A \cos B = \frac{1}{2}\sin(A-B) + \frac{1}{2}\sin(A+B)$$

$$\sin A \sin B = \frac{1}{2}\cos(A-B) - \frac{1}{2}\cos(A+B).$$

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Calculation of Fourier Coefficients

Theorem 1 Orthogonality Relations

If $j, k \in \mathbb{N}$, *then:*

$$\int_{-\pi}^{\pi} \cos jx \cos kx \, dx = \int_{-\pi}^{\pi} \sin jx \sin kx \, dx = \begin{cases} \pi & \text{if } j = k \\ 0 & \text{otherwise,} \end{cases}$$

and

$$\int_{-\pi}^{\pi} \sin jx \cos kx \, dx = 0.$$

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A Sketch of Proof

PROOF For $n \in \mathbb{Z}$, observe that

$$\int_{-\pi}^{\pi} \sin nx\, dx = 0 \qquad \text{and} \qquad \int_{-\pi}^{\pi} \cos nx\, dx = \begin{cases} 2\pi & \text{if } n = 0 \\ 0 & \text{otherwise.} \end{cases}$$

If $j, k \in \mathbb{N}$, we can use the product-to-sum identities to deduce that

$$\int_{-\pi}^{\pi} \cos jx \cos kx\, dx = \int_{-\pi}^{\pi} \frac{\cos(j-k)x + \cos(j+k)x}{2} dx = \begin{cases} \pi & \text{if } j = k \\ 0 & \text{otherwise,} \end{cases}$$

and

$$\int_{-\pi}^{\pi} \sin jx \sin kx\, dx = \int_{-\pi}^{\pi} \frac{\cos(j-k)x - \cos(j+k)x}{2} dx = \begin{cases} \pi & \text{if } j = k \\ 0 & \text{otherwise,} \end{cases}$$

and

$$\int_{-\pi}^{\pi} \sin jx \cos kx\, dx = \int_{-\pi}^{\pi} \frac{\sin(j-k)x + \sin(j+k)x}{2} dx = 0. \qquad \blacksquare$$

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Calculation of Fourier Coefficients

Theorem 2 Fourier Coefficients

Let

$$f(x) = a + \sum_{n=1}^{N} b_n \cos nx + \sum_{n=1}^{N} c_n \sin nx.$$

Then

$$a = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\, dx.$$

Furthermore, for all $n \in \{1, \ldots, N\}$,

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx\, dx \qquad \text{and} \qquad c_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx\, dx.$$

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Definition for f

Definition: Fourier Coefficients for f

The **Fourier coefficients** for a function $f : [-\pi, \pi] \to \mathbb{R}$ are the real number a and the sequences b_n and c_n defined by the following formulas:

$$a = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx, \quad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx, \quad c_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx.$$

Definition: Fourier Series for f

The **Fourier series** for a function $f : [-\pi, \pi] \to \mathbb{R}$ is the sum

$$a + \sum_{n=1}^{\infty} b_n \cos nx + \sum_{n=1}^{\infty} c_n \sin nx.$$

where a, b_n, and c_n are the Fourier coefficients for f.

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https://www.youtube.com/watch?v=PcYFnVBS_bg

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Summary

- ***Harmonics*** are sine waves that have frequencies at integer multiples of the fundamental frequency.
- When we ***add harmonics*** together, we create complex periodic patterns at the same fundamental frequency.
- A signal can be expressed as a sum of Sine Waves.

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